A Rejectable Causal Atlas for Experimental Archives: Composability, Honest Partial Identification, and Bridge Experiment Design

CAUSAL Lab Research Blueprint

Abstract

Modern experimentation is no longer scarce. Platform science, policy evaluation, and AI evaluation systems routinely produce archives containing many randomized or quasi-randomized contrasts, but these contrasts are usually stored as isolated point estimates. This paper asks when such an archive can be upgraded into a *causal atlas*: a structured map that composes old experiments only when their mechanisms, designs, and uncertainty certificates license transport to a target contrast; refuses composition when the archive lacks support; and selects bridge experiments that shrink the remaining unidentified region. The key object is therefore not semantic similarity between study descriptions, but a rejectable certificate of causal composability. We formalize each archive element as a treatment–outcome–context–design object endowed with an assumption profile, a mechanism representation, and an experiment-level debiased estimator. Under local smoothness, design compatibility, support, and moderator-sensitivity conditions, we prove a finite-sample composition inequality whose terms separate causal support error, curvature, statistical noise, nuisance-debiasing error, and hidden-moderator residuals. The construction retains the standard exact unbiasedness of Horvitz–Thompson/AIPW estimation under known randomization and carries an auxiliary finite-sample variance envelope into the atlas certificate; we further prove asymptotic normality and honest confidence intervals for atlas compositions, an impossibility theorem showing that semantic proximity alone cannot certify transport under hidden moderators, a valid partial-identification construction after rejection, a minimax lower bound showing that unsupported targets cannot be estimated faster than the certified gap permits, and a weak-submodular guarantee for greedy bridge design. Synthetic experiments with known mechanisms and a real-data NSW job-training local-contrast archive show that causal support improves reconstruction, restores interval coverage, and selects bridges that reduce unidentified regions more efficiently than semantic baselines.

1 Introduction

Causal ATLAS is a research program for turning experimental archives, LLM-generated interventions, routing logs, and strategic AI deployments into an identifiable, composable, and decision-aware causal map. This first paper supplies the archive-to-discovery layer of that program. Its central question is deliberately prior to prediction: *when is it mathematically legitimate to reuse one experiment as evidence about another?*

The broad field. Empirical science increasingly operates in an archival regime. A large digital platform may run thousands of A/B tests; a policy area may contain decades of randomized field experiments; an AI laboratory may evaluate prompts, model-routing choices, safety interventions, and user-interface treatments under partially overlapping tasks. The scientific promise is obvious: previous experiments should reduce the cost of new experimentation. The statistical danger is equally obvious: experiments that look similar in text, share a high-level topic, or have nearby embeddings need not identify the same causal effect. The broad field is therefore causal evidence synthesis: the attempt to combine randomized, quasi-randomized, and logged evidence without erasing the assumptions that made each piece of evidence credible.

The subproblem. Classical meta-analysis starts after a common estimand has been declared. Transportability theory asks whether an effect can be moved from one environment to another under structural assumptions. Representation-learning approaches retrieve studies by semantic or geometric proximity. The present paper studies the layer between retrieval and aggregation. Given an archive of experimental findings and a new target contrast, we ask whether the archive contains enough *causal support* to identify, approximate, or honestly bound the target effect. It is an identification problem over a finite and imperfect experimental archive.

A motivating failure. Consider a policy platform containing three apparently nearby interventions: a motivational email for job seekers, a peer-comparison message about job-search intensity, and a hiring-signal badge shown to employers. A language model may place all three close to a proposed target intervention, because the descriptions share vocabulary about encouragement, salience, or social proof. Yet their effects can depend on moderators not present in the text, such as baseline motivation or local labor-market tightness. They may also differ in experiment design: the treatment contrast, control-condition intensity, measurement scale, timing, exposure mapping, or institutional setting may not be aligned. These are distinct failure modes: moderators can change the treatment effect within a common causal scale, whereas design differences can place reported effects on different causal scales. A forced semantic composition may then report a confident effect for the target even though no archived design is directly comparable to the target contrast. In practice, the consequence is misallocated experimental budget or harmful policy transfer. In theory, the same failure appears as non-identification: two mechanisms can be arbitrarily close in the observed representation and still differ by a constant causal effect if a relevant moderator has been collapsed.

The proposed framework. We replace semantic support by a *rejectable causal-composition certificate*. Each experiment is represented as a treatment–outcome–context–design object with an uncertainty certificate and an assumption profile. A target is composed only after three checks: design compatibility, local mechanism support, and moderator sensitivity. If these checks pass, the atlas returns a debiased transported estimate and an honest interval whose radius decomposes into statistical and approximation terms. If they fail, the atlas returns a partial-identification interval and defines bridge value by the expected reduction in that interval's diameter. It then selects a new experiment targeted at the specific causal-support gap that prevented point identification. The framework is intentionally conservative in the only place where conservatism is scientific: it refuses point estimates when the archive lacks identifying support.

Why the contribution is not a sum of existing tools. A simple combination of meta-analysis, embeddings, and active learning would still miss the endogenous difficulty of this problem. Meta-analysis presumes comparability; embeddings propose similarity but do not identify causal effects; active learning typically values prediction accuracy rather than shrinkage of an unidentified causal set. The causal-atlas operation couples all three: semantic retrieval proposes candidates, causal support certifies or rejects composition, and bridge design is driven by the diameter of the partial-identification region. The technically nontrivial point is that the same support certificate controls finite-sample transport error, asymptotic inference, impossibility under hidden moderators, and minimax lower bounds. Thus rejection is not a heuristic filter. It is the statistical object that turns an archive into an identification device.

Main contributions.

1. **Causal experiment objects and assumptions with diagnostics.** We define an archive element as a treatment–outcome–context–design object carrying a target estimand, a design scale, an assumption profile, a mechanism representation, and an uncertainty certificate. The earlier semantic-atlas setting is recovered as the special case in which the representation is causally sufficient and all designs are compatible.

2. **Debiased archive estimators.** We replace raw plug-in effects by Horvitz–Thompson or AIPW experiment-level estimates. Their standard exact unbiasedness under known randomization is retained, while the cross-fitted product nuisance remainder and an auxiliary finite-sample variance envelope are carried explicitly into the causal-composition certificate.

3. **Finite-sample causal composition.** We prove an atlas composition inequality separating support residual, curvature, statistical noise, nuisance-debiasing error, and hidden-moderator residual. This gives a certificate rather than a score.

4. **Asymptotic inference.** We prove asymptotic normality for accepted atlas compositions and construct honest confidence intervals that remain valid when approximation bias is not negligible.

5. **Boundary results and minimax optimality.** We show that semantic proximity alone cannot certify causal transport under hidden moderators, and prove a minimax lower bound: unsupported targets incur an unavoidable error of the same order as the causal support gap, while statistical noise contributes the usual information-limited term.

6. **Partial identification and bridge design.** When composition fails, we construct a valid partial-identification interval. Its diameter becomes the value-of-information objective for bridge experiments, and a greedy bridge policy is near-optimal under weak submodularity.

7. **Evidence tied to theory.** Synthetic experiments with known mechanisms and a real NSW job-training local-contrast archive evaluate exactly the theoretical objects: reconstruction error, sign recovery, coverage, interval width, rejection, support sensitivity, ablations, and bridge shrinkage.

The paper is organized as follows. Section 2 positions the work relative to meta-analysis, transportability, partial identification, representation learning, and active design. Section 3 defines the mathematical objects and assumptions. Section 4 presents the estimator, the support test, and the bridge algorithm. Section 5 states the main finite-sample, asymptotic, impossibility, partial-identification, minimax, and bridge-design results. Section 6 reports synthetic and real-data evidence. Section 7 discusses scope boundaries and the larger Causal ATLAS roadmap. All proofs are collected in the appendix.

2 Related Work and Positioning

Evidence synthesis and meta-analysis. Meta-analysis and hierarchical modeling provide principled aggregation once a family of studies has been judged comparable. Random-effects models absorb heterogeneity through a parametric distribution, and Bayesian hierarchical models can borrow strength across related studies. These approaches solve an aggregation problem after the target estimand and exchangeability structure have already been accepted. Our problem is earlier. The atlas must decide whether the archive supports the target estimand at all. If not, the statistically correct output is not a random-effects mean but a set of compatible target effects.

Causal inference, external validity, and transportability. The language of potential outcomes (Neyman, 1923; Rubin, 1974; Imbens and Rubin, 2015) and structural causal models (Pearl, 2009) makes clear that each experiment answers a contrast under a design. Transportability and data fusion characterize when effects can be moved across environments under graphical and selection assumptions (Bareinboim and Pearl, 2016). We share the concern with external validity, but the object differs. Instead of transporting from one source to one target, we compose many archived contrasts through local convex support and design compatibility. This creates two geometric terms–support and curvature–that are absent from single-source transport formulas.

Debiased and orthogonal estimation. Robins–Rotnitzky–Zhao estimators and modern double/debiased machine learning use orthogonal scores to remove first-order nuisance bias (Robins et al., 1994; Chernozhukov et al., 2018). We use those ideas at the archive level. Each experiment contributes not merely a reported coefficient but a debiased effect estimate with a finite-sample uncertainty certificate. The atlas then composes these certified objects. This separation matters: a highly relevant experiment with an uncertified or biased estimator should not silently dominate a target reconstruction.

Partial identification and sensitivity analysis. Partial identification treats unidentifiable quantities as sets rather than points (Manski, 2003). Sensitivity analysis makes violations of assumptions explicit. Our contribution is to operationalize this logic inside an experimental archive. The diameter of the target's partial-identification set is both an uncertainty measure and an experimental-design objective. A gap is therefore not merely a warning; it is a measurable scientific opportunity.

Representation learning and experimental atlases. Language models and embeddings can organize large scientific archives. Recent experimental-atlas work uses semantic representations to link, reconcile, and bridge studies (Zhang et al., 2026). We adopt the atlas ambition but change the mathematical criterion. Text can retrieve candidates, summarize mechanisms, and propose bridges. It cannot, by itself, certify that all relevant moderators and design scales have been represented. Our impossibility theorem formalizes this boundary, while the semantic-special-case corollary states exactly when semantic composition becomes valid.

Active design and submodular optimization. Greedy selection under submodularity is a classical approximation principle (Nemhauser et al., 1978). In our setting, the objective is not prediction information or semantic novelty. It is expected shrinkage of a causal partial-identification interval. This distinction changes the selected experiment: the best bridge is the one that closes an identifying support gap, not necessarily the one whose description is most novel.

Positioning. The paper does not claim that text embeddings identify treatment effects, that all archived experiments are composable, or that bridge design automates scientific judgment. It claims a narrower and testable statement: if experiment objects carry enough mechanism, design, and uncertainty information to satisfy stated support and compatibility conditions, then local composition has a finite-sample and asymptotic error certificate; if those conditions fail, the archive should return partial identification and prioritize bridge experiments that reduce the certified gap.

3 Problem Setup

3.1 Experiment Objects, Samples, and Estimands

All random variables are defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, where Ω is the sample space, $\mathcal{F}$ is the associated σ-algebra, and $\mathbb{P}$ is the governing probability measure. For experiment i, $\mathbb{P}_i$ and $\mathbb{E}_i$ denote the probability law and expectation induced by its population, sampling frame, and design. The archive contains N experiment objects

$$\mathcal{E}_N = \{e_i : i \in [N]\}, \qquad e_i = (T_i, O_i, C_i, D_i, \mathcal{A}_i, r_i, \widehat{\tau}_i, \mathcal{U}_i),$$

where T_i is a treatment description, O_i an outcome definition, C_i a population-context descriptor, D_i the design metadata, $\mathcal{A}_i$ the assumption profile, and $r_i = r(e_i)$ the observed mechanism representation. The quantity $\widehat{\tau}_i$ is a debiased estimate of the causal effect, and

$$\mathcal{U}_i = (n_i, \widehat{v}_i, \mathfrak{b}_i)$$

is its uncertainty certificate, comprising the experiment sample size, the estimated influence-function variance, and a certified upper bound on the remaining nuisance-estimation bias. When experiments share units or logging systems, the archive additionally records the cross-experiment dependence certificate described in Assumption 3.4. The target object is

$$e_\star = (T_\star, O_\star, C_\star, D_\star, \mathcal{A}_\star, r_\star),$$

with unknown effect $\tau_\star$.

For each archive experiment i, let $W_{i\ell} = (X_{i\ell}, A_{i\ell}, Y_{i\ell})$, $\ell = 1, \ldots, n_i$, denote the unit-level record, where $A_{i\ell} \in \{0, 1\}$ is the assigned treatment or logged action. Let $Y_i(1), Y_i(0)$ be the potential outcomes on the normalized outcome scale recorded in D_i. The experiment-specific estimand is

$$\tau_i := \mathbb{E}_i\{Y_i(1) - Y_i(0)\}, \tag{3.1}$$

where $\mathbb{E}_i$ denotes expectation under the context and sampling frame of experiment i. For randomized experiments, (3.1) is the population average treatment effect in that experiment. For logged or quasi-randomized experiments, the notation is used only after $\mathcal{A}_i$ records the identifying assumptions, such as consistency, ignorability, overlap, exposure consistency, or negative-control completeness.

The target estimand is

$$\tau_\star := \mathbb{E}_\star\{Y_\star(1) - Y_\star(0)\}, \tag{3.2}$$

interpreted on the target design scale $(O_\star, D_\star)$. The atlas problem is to use $\mathcal{E}_N$ to produce either a certified estimate of $\tau_\star$ or an honest set of possible values.

3.2 Mechanism Space and Causal Support

Each experiment has a latent mechanism vector $m(e) \in \mathcal{M} \subset \mathbb{R}^p$. The coordinates may encode treatment semantics, outcome semantics, population context, design scale, control intensity, exposure mapping, and effect moderators. The observed representation is $r(e) \in \mathbb{R}^q$. In the ideal case

$r(e) = m(e)$. In the realistic case, $r(e)$ may omit or noisily proxy some moderator; this omission is represented by a sensitivity radius.

We assume the archive effects lie on an effect surface

$$\tau(e) = \mu\{m(e)\}, \qquad \mu : \mathcal{M} \to \mathbb{R}. \tag{3.3}$$

Write $m_i = m(e_i)$, $m_\star = m(e_\star)$, $r_i = r(e_i)$, and $r_\star = r(e_\star)$.

Definition 3.1 (Causal experiment object)**.** *An experiment object e is* causally encoded *if it specifies: (i) a target estimand $\tau(e)$; (ii) a mechanism representation $r(e)$ and a moderator-sensitivity certificate; (iii) a design profile D_e including treatment contrast, outcome scale, exposure mapping, control condition, and sampling frame; (iv) an assumption profile $\mathcal{A}_e$; and (v) a debiased estimate $\widehat{\tau}(e)$ together with its uncertainty certificate $\mathcal{U}(e)$.*

Definition 3.2 (Design compatibility)**.** *Let $\kappa(e, e') \in \{0, 1\}$ be a compatibility predicate. We set $\kappa(e, e') = 1$ when the treatment contrast, outcome scale, control condition, exposure mapping, sampling frame, and assumption profiles of e and e' can be placed on a common causal estimand scale after transformations explicitly recorded in $D_e, D_{e'}$ and $\mathcal{A}_e, \mathcal{A}_{e'}$. For $C \subseteq [N]$, write $\kappa(C, \star) = 1$ if $\kappa(e_j, e_\star) = 1$ for every $j \in C$.*

Definition 3.3 (Causal support and composability)**.** *For $C \subseteq [N]$, let $\Delta_C = \{\alpha \in \mathbb{R}_+^{|C|} : \sum_{j\in C} \alpha_j = 1\}$. For $\alpha \in \Delta_C$, define*

$$\bar{m}_\alpha = \sum_{j\in C} \alpha_j m_j, \qquad \rho_\star(\alpha) = \|m_\star - \bar{m}_\alpha\|.$$

The target is δ-causally composable from C if $\kappa(C, \star) = 1$ and there exists $\alpha \in \Delta_C$ such that $\rho_\star(\alpha) \le \delta$. The quantity $\rho_\star(\alpha)$ is the causal support residual.

This definition separates three objects that are often conflated. Semantic proximity is closeness of observed descriptions. Causal support is closeness in the mechanism space relevant to μ. Design compatibility is the statement that the effects being combined live on a common causal scale.

3.3 Assumptions and Their Scientific Roles

Assumption 3.1 (Identified experiment-level effects)**.** *For each archive experiment i, the unit records are independent conditional on the experiment design. Consistency holds, and treatment is assigned or logged with propensity $\pi_i(x) = \mathbb{P}_i(A = 1 \mid X = x)$ satisfying $\pi_0 \le \pi_i(x) \le 1 - \pi_0$ for a fixed $\pi_0 \in (0, 1/2)$. Conditional ignorability holds whenever the experiment is not literally randomized:*

$$(Y_i(1), Y_i(0)) \perp\!\!\!\perp A_i \mid X_i.$$

The analyst either knows π_i from the design/logging policy or supplies a cross-fitted estimate whose error is explicitly recorded.

Role. This assumption identifies each archive contrast before any atlas composition is attempted. *Diagnosis.* Randomization logs, balance checks, overlap plots, and design audits are the empirical diagnostics. *Fallback.* If ignorability or overlap is doubtful, the object remains in the archive but is used only through sensitivity bounds or negative-control certificates.

Assumption 3.2 (Design compatibility and normalization)**.** *If $\kappa(C, \star) = 1$, then all effects $\{\tau_j : j \in C\}$ and $\tau_\star$ are expressed on a common normalized causal scale. Any scale transformation is deterministic, recorded in $D_j, D_\star$, and has been applied before composition.*

Role. This prevents the atlas from adding incomparable quantities, such as short-run click effects and long-run earnings effects. *Diagnosis.* The predicate is checked from outcome definitions, treatment contrasts, exposure mappings, and sampling frames. *Fallback.* Incompatible studies may still be used for qualitative retrieval or for proposing bridges, but not for point composition.

Assumption 3.3 (Local smoothness of the effect surface)**.** *The mechanism set $\mathcal{M} \subset \mathbb{R}^p$ is compact and convex. The effect surface $\mu : \mathcal{M} \to \mathbb{R}$ is continuously differentiable with $\|\nabla\mu(m)\| \leq L$ for all $m \in \mathcal{M}$, and its gradient is H-Lipschitz:*

$$\left\|\nabla\mu(m) - \nabla\mu(m')\right\| \leq H \left\|m - m'\right\| \qquad \text{for all } m, m' \in \mathcal{M}.$$

Role. Smoothness turns local mechanism support into an approximation guarantee. *Diagnosis.* It is stress-tested by negative controls, moderator curves, residual-versus-distance diagnostics, and leave-one-region-out validation. *Fallback.* If only Lipschitz continuity is credible, the curvature term is replaced by a coarser nearest-neighbor bound and intervals widen.

Assumption 3.4 (Uncertainty certificates)**.** *After debiasing, write $\widehat{\tau}_i = \tau_i + \xi_i + b_i$, where b_i is an explicit nuisance-bias remainder and ξ_i is conditionally mean-zero. Conditional on the experiment designs and nuisance-training folds, ξ_i is sub-Gaussian with variance proxy $s_i^2 = v_i/n_i$:*

$$\mathbb{E}\{\exp(\lambda\xi_i) \mid D_i, C_i, \mathcal{A}_i\} \leq \exp(\lambda^2 s_i^2/2), \qquad \lambda \in \mathbb{R}.$$

For independently run experiments, ξ_i's are conditionally independent. If experiments share units or logging systems, the analyst supplies a sub-Gaussian proxy-matrix certificate Σ and the term $\sum_j \alpha_j^2 s_j^2$ below is replaced by $\alpha^\top \Sigma \alpha$.

Role. The certificate separates statistical noise from support failure. *Diagnosis.* It is checked through randomization variance, sandwich estimates, bootstrap stability, or influence-function diagnostics. *Fallback.* Heavy-tailed outcomes can be handled by robust mean estimators or empirical Bernstein bounds.

Assumption 3.5 (Observed representation and moderator certificate)**.** *For every design-compatible weighted composition α, the analyst supplies a nonnegative hidden-moderator residual $R_{\text{hid}}(\alpha)$ such that*

$$R_{\text{hid}}(\alpha) \geq \left| \mu(m_\star) - \mu(\widetilde{m}_\star) - \sum_{j \in C} \alpha_j \{\mu(m_j) - \mu(\widetilde{m}_j)\} \right|$$

for all mechanisms $\widetilde{m}_e$ compatible with the observed representation and the stated sensitivity radius. In the fully observed case, $R_{\text{hid}}(\alpha) = 0$.

Role. This is the formal place where omitted moderators enter. *Diagnosis.* The radius is informed by measured moderator proxies, sensitivity curves, expert coding, and external validation. *Fallback.* If no credible radius can be supplied, the atlas refuses point transport and reports partial identification.

4 Method: Debiased Estimation and a Rejectable Causal Atlas

The method has four steps: debias each archive effect, retrieve candidates, test causal support and design compatibility, and either transport the target effect or design a bridge.

4.1 Experiment-Level Debiasing

For each experiment i, define the nuisance functions

$$\mu_{i,a}(x) = \mathbb{E}_i(Y_i \mid A_i = a, X_i = x), \qquad a \in \{0, 1\}.$$

Given cross-fitted nuisance estimates $\widehat{\eta}_i = (\widehat{\mu}_{i,0}, \widehat{\mu}_{i,1}, \widehat{\pi}_i)$, the AIPW score is

$$\phi_i(W; \widehat{\eta}_i) = \widehat{\mu}_{i,1}(X) - \widehat{\mu}_{i,0}(X) + \frac{A\{Y - \widehat{\mu}_{i,1}(X)\}}{\widehat{\pi}_i(X)} - \frac{(1-A)\{Y - \widehat{\mu}_{i,0}(X)\}}{1 - \widehat{\pi}_i(X)}. \tag{4.1}$$

The archive estimate and its variance component are

$$\widehat{\tau}_i = \frac{1}{n_i} \sum_{\ell=1}^{n_i} \phi_i(W_{i\ell}; \widehat{\eta}_i^{(-\ell)}), \qquad \widehat{v}_i = \frac{1}{n_i - 1} \sum_{\ell=1}^{n_i} \left\{ \phi_i(W_{i\ell}; \widehat{\eta}_i^{(-\ell)}) - \widehat{\tau}_i \right\}^2, \tag{4.2}$$

where $\widehat{\eta}_i^{(-\ell)}$ denotes a nuisance fit trained without unit ℓ's fold. In a randomized experiment with known propensity, $\widehat{\pi}_i = \pi_i$; each evaluation-fold average is conditionally unbiased, and therefore (4.2) is unbiased after averaging over folds. If no covariates are used, it reduces to the Horvitz–Thompson difference-in-means estimator.

When the propensity is estimated, the uncertainty certificate stores $|b_i| \le \mathfrak{b}_i$, where the cross-fitted product-error bound is

$$\mathfrak{b}_i = \sum_{k=1}^{K} \frac{|I_{ik}|}{n_i} \mathbb{E} \left[\frac{\left\| \widehat{\pi}_i^{(-k)} - \pi_i \right\|_{2,i}}{\pi_0} \left\{ \left\| \widehat{\mu}_{i,1}^{(-k)} - \mu_{i,1} \right\|_{2,i} + \left\| \widehat{\mu}_{i,0}^{(-k)} - \mu_{i,0} \right\|_{2,i} \right\} \right]. \tag{4.3}$$

This is a standard AIPW orthogonality calculation; the main theoretical contribution lies in carrying its bound into the composition radius. It is developed fully in the Appendix. Appendix A.2 gives the full conditional-expectation and Cauchy–Schwarz derivation because $\mathfrak{b}_i$ enters the composition radius.

4.2 Support Testing and Transport

For a target $e_\star$, the atlas first retrieves a candidate set C using semantic and metadata filters. Retrieval is deliberately permissive. Identification is performed by the next layer. In implementation, every experiment is assigned a versioned design signature containing the treatment contrast and control condition, outcome definition and units, exposure mapping, sampling frame and time horizon, and the assumptions needed to identify its stored effect. The compatibility check is applied before outcomes enter the weighting program. We set $\kappa(e_j, e_\star) = 1$ only when each required field either matches the target or has an analyst-approved deterministic map to the target scale; that map is applied to $\widehat{\tau}_j$, $\widehat{v}_j$, and the nuisance-bias certificate before composition. Missing required metadata, an unverified scale conversion, or contradictory assumption profiles are fail-closed conditions: the candidate is removed and the reason is stored in the audit record. If no compatible candidate remains, the support optimization is skipped and the target is routed directly to partial identification and bridge design.

After this deterministic screening, let C denote the retained set. The atlas solves

$$\widehat{\alpha} \in \operatorname*{arg\,min}_{\alpha \in \Delta_C} \left\{ \left\| r_\star - \sum_{j \in C} \alpha_j r_j \right\|^2 + \lambda_\sigma \sum_{j \in C} \alpha_j^2 \frac{\widehat{v}_j}{n_j} + \lambda_{\text{hid}} \widehat{R}_{\text{hid}}(\alpha) \right\}. \tag{4.4}$$

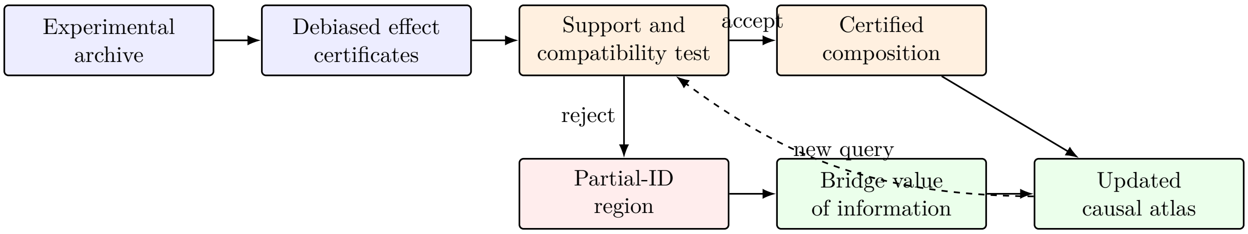

Figure 1: A causal atlas is deliberately rejectable. Semantic retrieval proposes candidate experiments; debiased uncertainty certificates, design compatibility, support, and moderator sensitivity decide whether composition is licensed. Rejection produces a partial-identification region and a bridge-design objective rather than a forced point estimate.

The first term measures observed support, the second discourages concentration on noisy experiments, and the third penalizes sensitivity to omitted moderators. Composition is accepted only if the certificate radius implied by Theorem 5.1 is below a scientific tolerance. The transported estimator is

$$\widehat{\theta}_\alpha = \sum_{j \in C} \alpha_j \widehat{\tau}_j. \tag{4.5}$$

4.3 Bridge Experiment Design

Once the support certificate is too large for point release, the method returns a partial-identification region and uses its diameter to decide which experiment should be run next. Let $\Theta_A(e_\star)$ denote the set of target effects compatible with archive A and the stated identifying and sensitivity restrictions, and let $\widehat{\Theta}_A(e_\star)$ denote its certified sample analogue.

Definition 4.1 (Bridge value of information)**.** *For candidate bridge experiment b, define*

$$\mathrm{VoI}_A(b) = \mathrm{diam}\{\Theta_A(e_\star)\} - \mathbb{E}_b\big[\,\mathrm{diam}\{\Theta_{A\cup\{b\}}(e_\star)\}\big],$$

where the expectation is over possible bridge outcomes under the proposed design.

In the operational implementation, the unknown regions Θ_A are replaced by their certified estimates $\widehat{\Theta}_A$, producing an estimated marginal value $\widehat{\mathrm{VoI}}(u \mid S)$ conditional on the bridges already selected in S. Starting from $S_0 = \varnothing$, the planner selects

$$u_b \in \operatorname*{arg\,max}_{u \in \mathcal{B} \setminus S_{b-1}} \widehat{\mathrm{VoI}}(u \mid S_{b-1}), \qquad S_b = S_{b-1} \cup \{u_b\}, \qquad b = 1, \dots, B.$$

After each proposed bridge is added, its estimated effect and uncertainty certificate are incorporated into the archive, the partial-identification region is recomputed, and the remaining marginal values are updated. Thus, the bridge rule targets expected shrinkage of the unidentified region rather than semantic novelty. The approximation guarantee for this greedy rule is stated in the Main Results section.

Algorithm 1 Rejectable Causal Atlas and Greedy Bridge Design

Require: Archive $\mathcal{E}_N$; target object $e_\star$; candidate bridge library $\mathcal{B}$; budget B; support tolerance δ_{sup}; smoothness constants L, H; confidence level $1 - \zeta$.

Ensure: Certified estimate, partial-identification interval, and/or selected bridge set.

Debias each archive effect using (4.2); store $(r_i, D_i, \mathcal{A}_i, \widehat{\tau}_i, \mathcal{U}_i)$.
Retrieve candidate set $C \subseteq [N]$ by semantic and metadata screening.
Build the design signature for each candidate; apply every approved scale map to its estimate and uncertainty certificate.
Remove every $j \in C$ with $\kappa(e_j, e_\star) = 0$, recording the failed field or missing map.
if $C = \varnothing$ **then**
  return rejection with the broad target-effect bounds encoded by $\mathcal{F}_A$, and invoke bridge design.
end if
Compute $\widehat{\alpha}$ by the support program (4.4).
Compute the certificate radius $Rad(\widehat{\alpha}, \zeta)$ from Theorem 5.1.
if support residual, nuisance remainder, and moderator sensitivity imply $Rad(\widehat{\alpha}, \zeta) \leq \delta_{\text{sup}}$ **then**
  return $\widehat{\theta}_{\widehat{\alpha}} = \sum_{j \in C} \widehat{\alpha}_j \widehat{\tau}_j$ and the honest interval in Theorem 5.2.
else
  Construct $\widehat{\Theta}_A(e_\star)$ from Theorem 5.4.
  Initialize $S_0 = \varnothing$.
  for $b = 1, \ldots, B$ **do**
    Select $u_b \in \arg\max_{u \in \mathcal{B} \setminus S_{b-1}} \widehat{\mathrm{VoI}}(u \mid S_{b-1})$.
    Set $S_b = S_{b-1} \cup \{u_b\}$.
  end for
  return partial-identification interval $\widehat{\Theta}_A(e_\star)$ and bridge set S_B.
end if

5 Main Results

The results are organized to mirror the operational logic of the causal atlas. Section 5 first establishes the validity of the experiment-level evidence, then determines when those certified effects can be composed, and finally characterizes what the atlas should report when composition fails.

The first subsection, *Debiased Archive Estimation*, records only the auxiliary variance envelope used by the composition analysis. The standard AIPW unbiasedness and nuisance-remainder calculations are part of the method in Section 4 and are derived in Appendix A.2, without a separate proposition.

The second subsection, *Finite-Sample Composition*, establishes the central transport certificate. Lemma 5.2 shows that the intersection of compatible certified intervals is a well-defined interval whenever the certificates are mutually consistent. Theorem 5.1 then bounds the error of a transported composition by separating causal-support error, local curvature, hidden-moderator sensitivity, nuisance bias, and statistical noise. The familiar fully observed affine identity is recorded as a boundary case within this theorem.

The third subsection, *Asymptotic Normality and Honest Confidence Intervals*, establishes the large-sample inferential behavior of an accepted composition. Theorem 5.2 gives both asymptotic normality and the resulting confidence interval, explicitly inflating the interval by the certified deterministic radius when approximation error is non-negligible.

The fourth subsection, *Semantic Failure and Partial Identification*, establishes the boundary of semantic retrieval and the valid fallback after rejection. Theorem 5.3 proves that observed-representation proximity alone cannot uniformly certify causal transport when relevant moderators are hidden. The archive-induced partial-identification region and Theorem 5.4 then show that failed point composition can still produce a coverage-valid set of target effects when point composition lacks support.

The fifth subsection, *Minimax Lower Bound*, establishes that this conservatism is information-theoretically necessary. Theorem 5.5 proves that unsupported targets incur an unavoidable geometric error, while finite archive noise contributes an additional information-limited term. Thus, the error cannot be removed by replacing the atlas with a more sophisticated weighting algorithm alone.

The final subsection, *Bridge Experiment Design*, turns the remaining unidentified region into an experimental-design objective. The bridge value-of-information definition measures the expected reduction in the partial-identification diameter, and Theorem 5.6 proves that greedy selection is near-optimal when this objective is monotone and weakly submodular, allowing for uniformly estimated marginal-value errors.

5.1 Debiased Archive Estimation

Lemma 5.1 (Archive variance envelope)**.** *Suppose Assumption 3.1 holds, the nuisance estimates are cross-fitted with $\widehat{\pi}_i \in [\pi_0, 1-\pi_0]$, $|Y_i| \le M$, and $\widehat{\mu}_{i,a} \in [-M, M]$. If the evaluation scores are conditionally independent given a common external nuisance-training sample, then for independent archive experiments and any $\alpha \in \Delta_C$,*

$$\operatorname{Var}\left(\sum_{j\in C} \alpha_j \widehat{\tau}_j \,\middle|\, \widehat{\eta}\right) \le \sum_{j\in C} \alpha_j^2 \frac{C_M^2}{n_j}, \qquad C_M = 2M + \frac{2M}{\pi_0}.$$

For ordinary fold-specific cross-fitting, the operational analogue is the uncertainty certificate in Assumption 3.4.

Thus the archive stores a debiased effect together with its nuisance and variance certificates; the standard experiment-level facts serve the later composition theorem without being presented as contributions of their own.

5.2 Finite-Sample Composition

For $\alpha \in \Delta_C$, define the deterministic approximation radius

$$D(\alpha) = L\rho_\star(\alpha) + \frac{H}{2}\sum_{j\in C} \alpha_j \left\|m_j - \bar{m}_\alpha\right\|^2 + R_{\mathrm{hid}}(\alpha) + \sum_{j\in C} \alpha_j \mathfrak{b}_j. \tag{5.1}$$

The four terms are, respectively, causal support error, local curvature, hidden-moderator sensitivity, and nuisance-debiasing remainder.

Lemma 5.2 (Well-posed atlas interval)**.** *Let C be finite and let $\mathcal{W}_A(e_\star) \subseteq \Delta_C$ be a nonempty feasible weight set. For any nonnegative radius function $B : \mathcal{W}_A(e_\star) \to [0, \infty)$, the set*

$$\widehat{\Theta}_A(e_\star) = \bigcap_{\alpha\in\mathcal{W}_A(e_\star)} \left[\sum_{j\in C} \alpha_j \widehat{\tau}_j - B(\alpha), \sum_{j\in C} \alpha_j \widehat{\tau}_j + B(\alpha)\right]$$

is a closed interval whenever it is nonempty. If it is empty, the archive estimates and the stated certificates are mutually inconsistent at the chosen confidence level.

Theorem 5.1 (Finite-sample causal composition bound)**.** *Let $C \subseteq [N]$ satisfy $\kappa(C, \star) = 1$, and fix $\alpha \in \Delta_C$. Under Assumptions 3.2–3.5, with probability at least $1 - \zeta$,*

$$\begin{aligned}\left|\tau_\star - \sum_{j\in C} \alpha_j \widehat{\tau}_j\right| \le L\rho_\star(\alpha) + \frac{H}{2}\sum_{j\in C}\alpha_j \|m_j - \bar{m}_\alpha\|^2 + R_{\text{hid}}(\alpha) + \sum_{j\in C}\alpha_j \mathfrak{b}_j \\ + \sqrt{2\log(2/\zeta)\sum_{j\in C}\alpha_j^2 s_j^2} \equiv \mathit{Rad}(\alpha).\end{aligned} \tag{5.2}$$

If the archive experiments have a certified sub-Gaussian proxy matrix Σ under correlated archive errors, use $\sqrt{2\log(2/\zeta)\alpha^\top \Sigma\alpha}$.

Scientific reading. The theorem says that reuse is valid only when five quantities are controlled. The target must lie in causal support, the effect surface must be nearly linear over the local hull, the archived estimates must be precise, the debiasing remainder must be small, and omitted moderators must be certified. It does not say that semantic neighbors are valid; semantic retrieval merely supplies candidates.

Affine boundary case. The familiar semantic-atlas identity follows directly from Theorem 5.1, so we do not elevate it to a separate result. If $r(e) = m(e)$, $R_{\text{hid}}(\alpha) = 0$, $\mathfrak{b}_j = 0$, $\kappa(C, \star) = 1$, μ is affine on the relevant convex hull, and $m_\star = \sum_{j\in C}\alpha_j m_j$, then

$$\tau_\star = \sum_{j\in C}\alpha_j \tau_j.$$

With estimated archive effects, the only remaining term in (5.2) is statistical. This boundary case clarifies why semantic composition can work under causal sufficiency without presenting the classical affine identity as a standalone contribution.

5.3 Asymptotic Normality and Honest Confidence Intervals

Let $\theta_\alpha = \sum_{j\in C}\alpha_j\tau_j$ and $V_{\alpha,n} = \sum_{j\in C}\alpha_j^2 v_j/n_j$. The next theorem separates statistical normality around the composed archive estimand θ_α from approximation bias relative to $\tau_\star$.

Theorem 5.2 (Asymptotic distribution and honest interval for an accepted composition)**.** *Consider a triangular sequence of archives with candidate sets C_n and weights α_n. Suppose Assumptions 3.1–3.5 hold. Write*

$$\psi_{j\ell} = \phi_j(W_{j\ell};\eta_j) - \tau_j, \qquad v_j = \mathrm{Var}(\psi_{j\ell}),$$

and suppose the cross-fitted estimators admit the aggregate asymptotic linear expansion

$$\widehat{\tau}_j - \tau_j = \frac{1}{n_j}\sum_{\ell=1}^{n_j}\psi_{j\ell} + r_{j,n}, \qquad \sum_{j\in C_n}\alpha_{n,j} r_{j,n} = o_{\mathbb{P}}(V_{\alpha,n}^{1/2}),$$

where

$$V_{\alpha,n} = \sum_{j\in C_n}\alpha_{n,j}^2\frac{v_j}{n_j}.$$

This remainder condition is the precise aggregate nuisance-rate requirement used in the proof; it is typically obtained from cross-fitting and the product-rate calculation in Appendix A.2. Assume the independent influence-function summands satisfy the Lindeberg condition

$$\frac{1}{V_{\alpha,n}} \sum_{j \in C_n} \frac{\alpha_{n,j}^2}{n_j} \mathbb{E}\left[\{\phi_j(W_j;\eta_j) - \tau_j\}^2 \mathbf{1}\left\{|\alpha_{n,j}(\phi_j(W_j;\eta_j) - \tau_j)/n_j| > \varepsilon V_{\alpha,n}^{1/2}\right\}\right] \to 0$$

for every $\varepsilon > 0$. Then

$$\frac{\widehat{\theta}_{\alpha_n} - \theta_{\alpha_n}}{V_{\alpha,n}^{1/2}} \rightsquigarrow \mathsf{N}(0,1). \tag{5.3}$$

If $\widehat{V}_{\alpha,n}/V_{\alpha,n} \to_{\mathbb{P}} 1$, the studentized statistic is also asymptotically standard normal. Moreover, if the deterministic approximation radius $D(\alpha_n) = o(V_{\alpha,n}^{1/2})$, then

$$\frac{\widehat{\theta}_{\alpha_n} - \tau_\star}{V_{\alpha,n}^{1/2}} \rightsquigarrow \mathsf{N}(0,1).$$

Suppressing the sequence index in the following display, and without requiring $D(\alpha_n) = o(V_{\alpha,n}^{1/2})$, define

$$\mathrm{CI}_{1-\zeta}(\alpha) = \left[\widehat{\theta}_\alpha - z_{1-\zeta/2}\widehat{V}_\alpha^{1/2} - \widehat{D}(\alpha), \widehat{\theta}_\alpha + z_{1-\zeta/2}\widehat{V}_\alpha^{1/2} + \widehat{D}(\alpha)\right] \tag{5.4}$$

This interval has asymptotic coverage at least $1-\zeta$ whenever the one-sided radius underestimation is negligible:

$$\frac{\{D(\alpha) - \widehat{D}(\alpha)\}_+}{V_\alpha^{1/2}} \xrightarrow{\mathbb{P}} 0.$$

This condition permits conservative overestimation of $D(\alpha)$ and rules out only first-order underestimation. If $D(\alpha) = o(V_\alpha^{1/2})$, the shorter Wald interval without $\widehat{D}(\alpha)$ is valid.

This is the inferential boundary of the atlas. A target in sufficiently good causal support receives an ordinary asymptotic confidence interval. A target with nonnegligible support or moderator error receives an honest interval inflated by $\widehat{D}(\alpha)$. A target whose inflated interval is too wide is routed to partial identification and bridge design.

5.4 Semantic Failure and Partial Identification

Theorem 5.3 (Semantic proximity is insufficient under hidden moderators)**.** *Let $x(e)$ be an observed representation for which there exist two admissible experiments e^0, e^1 with $x(e^0) = x(e^1)$ but $h(e^0) \neq h(e^1)$ for a moderator coordinate h allowed by the model class. Then for every $\eta > 0$, there exists an effect surface μ satisfying Assumption 3.3 and two targets with $\|x(e^0) - x(e^1)\| \leq \eta$ but*

$$\left|\tau(e^1) - \tau(e^0)\right| \geq c$$

for a constant $c > 0$ determined by the moderator range and the Lipschitz radius. Hence no procedure based only on x-distance can uniformly certify causal composability over this model class.

A common objection is that a sufficiently strong embedding could include the moderator. That objection is correct and is precisely why the theorem is useful. If the representation is causally sufficient, the theorem does not apply and the affine boundary case following Theorem 5.1 describes the favorable setting. The impossibility concerns representations that are many-to-one over moderators relevant to treatment effects.

Definition 5.1 (Archive-induced partial-identification region)**.** *Let $\mathcal{F}_A$ be the class of effect surfaces satisfying the stated smoothness, design-compatibility, moderator, and uncertainty certificates. The archive-induced partial-identification region for the target is*

$$\Theta_A(e_\star) = \{f(m_\star) : f \in \mathcal{F}_A\}.$$

Theorem 5.4 (Valid partial identification from failed composition)**.** *Assume the conditions of Theorem 5.1. Let $\mathcal{W}_A(e_\star)$ be a finite collection of design-compatible weights and choose confidence levels $\{\zeta_\alpha : \alpha \in \mathcal{W}_A(e_\star)\}$ with $\sum_{\alpha\in\mathcal{W}_A(e_\star)} \zeta_\alpha \le \zeta$. Define*

$$\widehat{\Theta}_A(e_\star) = \bigcap_{\alpha\in\mathcal{W}_A(e_\star)} \left[\sum_j \alpha_j \widehat{\tau}_j - Rad(\alpha, \zeta_\alpha), \sum_j \alpha_j \widehat{\tau}_j + Rad(\alpha, \zeta_\alpha)\right], \tag{5.5}$$

where $Rad(\alpha, u)$ denotes the right-hand side of (5.2) with confidence argument u. Then

$$\mathbb{P}\{\tau_\star \in \widehat{\Theta}_A(e_\star)\} \ge 1 - \zeta.$$

Moreover, if every design-compatible convex combination has support residual at least $r_0 > 0$, then point identification is impossible over the Lipschitz class without an additional restriction: there exist two admissible effect surfaces agreeing on the archive and differing at $m_\star$ by at least $c_0 \min\{Lr_0, M\}$, where M is the global effect bound and $c_0 > 0$ is universal.

Failed composition is therefore not a failed method. It is a certificate that the archive does not contain the target mechanism at the requested resolution. The width of $\widehat{\Theta}_A(e_\star)$ quantifies the missing bridge.

5.5 Minimax Lower Bound

The previous theorem states non-identification. The next result states an information-theoretic lower bound. It shows that the atlas certificate is not merely conservative: its support and variance terms are unavoidable in the worst case.

Theorem 5.5 (Minimax lower bound for unsupported targets)**.** *Fix a design-compatible archive hull $K = \mathrm{conv}\{m_j : j \in C, \kappa(e_j, e_\star) = 1\}$ and let $d_\star = \mathrm{dist}(m_\star, K)$. Consider the Gaussian archive experiment*

$$Z_j = \mu(m_j) + \varepsilon_j, \qquad \varepsilon_j \perp\!\!\!\perp \mathsf{N}(0, s_j^2), \qquad j \in C,$$

with known s_j^2. Let $\mathcal{F}(L, M)$ be the class of functions $\mu : \mathcal{M} \to [-M, M]$ that are L-Lipschitz. Then for a universal constant $c > 0$,

$$\inf_{\widehat{T}} \sup_{\mu\in\mathcal{F}(L,M)} \mathbb{E}_\mu \left|\widehat{T} - \mu(m_\star)\right| \ge c \left[\{Ld_\star\} \wedge M \ \vee \ \left\{\sum_{j\in C} s_j^{-2}\right\}^{-1/2} \wedge M\right]. \tag{5.6}$$

If $d_\star > 0$, no estimator can uniformly remove the geometric support error without adding assumptions beyond $\mathcal{F}(L, M)$. If $d_\star = 0$, the remaining lower bound is the usual Fisher-information term for estimating a common local level.

The lower bound explains why the atlas reports partial identification when the target lies outside the design-compatible hull. When the target lies outside the design-compatible hull, the error floor is geometric, not computational. Better algorithms cannot recover an unsupported effect; only stronger assumptions or a bridge experiment can.

5.6 Greedy Bridge-Design Guarantee

The Method section defines the bridge value of information and the adaptive greedy rule. This subsection establishes when that rule is close to the best policy available in the finite bridge library.

Theorem 5.6 (Greedy adaptive bridge-design guarantee)**.** *Let $\mathcal{B}$ be a finite bridge library. For $S \subseteq \mathcal{B}$, write*

$$F(S) = \operatorname{diam}\{\Theta_A(e_\star)\} - \mathbb{E}[\operatorname{diam}\{\Theta_{A\cup S}(e_\star)\}].$$

Assume F is monotone, $F(\varnothing) = 0$, and γ-weakly submodular:

$$\sum_{b\in T\setminus S} \{F(S\cup\{b\}) - F(S)\} \geq \gamma\{F(T) - F(S)\}$$

for all $S \subseteq T \subseteq \mathcal{B}$. Suppose greedy bridge design selects elements using estimated marginal values whose uniform absolute error is at most ε_{est}. If S_B is the greedy set after B bridges and $S^\star \in \arg\max_{|S|\leq B} F(S)$, then

$$F(S_B) \geq (1 - e^{-\gamma})F(S^\star) - \frac{2B\varepsilon_{\text{est}}}{\gamma}. \tag{5.7}$$

Equivalently,

$$\mathbb{E}[\operatorname{diam}\{\Theta_{A\cup S_B}(e_\star)\}] \leq \operatorname{diam}\{\Theta_A(e_\star)\} - (1 - e^{-\gamma})F(S^\star) + \frac{2B\varepsilon_{\text{est}}}{\gamma}.$$

The guarantee applies to bridge objectives with approximate diminishing returns; under that condition, a greedy atlas policy is near-optimal for shrinking the unidentified region. This is why the empirical section evaluates bridge selection by partial-identification diameter, the design objective.

6 Experiments: Synthetic and Real-World Analysis

The experiments are designed to test the theory rather than to decorate it. Each reported table corresponds to a theoretical object. Reconstruction error tests the composition bound; coverage tests uncertainty calibration; rejection rate tests support diagnostics; interval width tests partial identification; bridge shrinkage tests Theorem 5.6; sensitivity curves stress Assumption 3.5.

6.1 Experimental Setup

Synthetic data-generating process. Each synthetic archive contains eight randomized experiments. The mechanism is $m = (s_1, s_2, h, q) \in [-1, 1]^4$, where s_1 and s_2 are public semantic coordinates, h is a hidden moderator observed through a bounded proxy, and q records a design/context coordinate. A nonlinear effect surface with interactions and curvature generates the experiment-level effects. Every nominal experiment contains 400 units with treatment probability 0.5; the target also contains 400 units and is used only to supply the synthetic truth after prediction. This construction permits semantic neighbors to be causally incompatible through variation in h and q. The complete surface, proxy construction, analytic bounds, and optimizer constants are given in Appendix B.1.

Methods and baselines. The proposed Causal ATLAS uses the full public causal representation, design compatibility, estimated effect uncertainty, and the declared moderator certificate. We compare it with the same composition forced to release every target, semantic forced composition using only (s_1, s_2), nearest semantic retrieval, and a global archive mean. The shared-target benchmark additionally includes an evaluation-only oracle latent-support reference: it replaces the public representation by the simulated latent mechanism, but it is unavailable to any deployable procedure and is never used for retrieval, weighting, release, or bridge selection. Complete formal ablations for removing the variance penalty and restricting retrieval to four candidates are reported in Appendix B.2 (B.2).

Metrics and protocol. We pool three prespecified seed batches and 100 target draws per seed. All methods receive the same archive–target draw whenever they are compared. We report release rate, MAE, RMSE, and sign accuracy. In the synthetic benchmarks, point-error metrics for a rejectable method are conditioned on release; in the NSW reconstruction stress test, we additionally retain all-target raw point metrics to separate point-estimator behavior from the release rule. Interval coverage is always reported jointly with mean width. For a released subset $\mathcal{R}$, the selective risk is

$$\mathrm{MAE}_{\text{released}} = \mathbb{E}[|\widehat{\tau} - \tau^{\star}| \mid \text{release}]\,,$$

which is not an unconditional accuracy claim. Partial-identification diameter is evaluated on rejected targets, and bridge quality is the reduction in that same diameter. The repository retains seed-wise summaries and Monte Carlo errors; the main section emphasizes effect sizes and the behavior of the decision rule.

6.2 Certified Causal Composition

The shared-target benchmark shows that causal composition and abstention work together. Causal ATLAS releases 0.463 of the 300 common targets and has released-target MAE 0.111, whereas forcing the same weights to publish all targets gives MAE 0.139 (Table 1). The oracle latent-support reference has MAE 0.135 on the full target population. The released ATLAS subset and the full-population oracle are therefore not directly comparable as unconditional risks; the fair representation comparison is between ATLAS with and without rejection. Semantic forced, nearest semantic, and global mean have MAEs 0.249, 0.553, and 0.282, respectively.

Table 1: Shared-target synthetic benchmark on the same 300 target draws. The evaluation population is explicit: Causal ATLAS point metrics are conditional on its released targets (139 of 300), whereas ATLAS without rejection, all forced baselines, and the evaluation-only latent oracle are evaluated on all 300 targets. The released ATLAS MAE is therefore not directly comparable with the all-target oracle MAE.

Method	Evaluation population	Release	MAE	RMSE	Sign	Coverage	Width
Causal ATLAS	Released targets (139/300)	0.463	0.111	0.138	0.928	1.000	2.946
ATLAS, no rejection	All targets (300/300)	1.000	0.139	0.170	0.893	1.000	3.339
Semantic forced	All targets (300/300)	1.000	0.249	0.310	0.797	1.000	2.831
Nearest semantic	All targets (300/300)	1.000	0.553	0.697	0.717	0.970	3.127
Global mean	All targets (300/300)	1.000	0.282	0.362	0.780	1.000	3.564
Oracle latent support†	All targets (300/300)	1.000	0.135	0.160	0.893	1.000	3.337

† The oracle replaces the public representation with simulated latent mechanisms and is evaluation-only; it is unavailable to retrieval, weighting, release, or bridge selection.

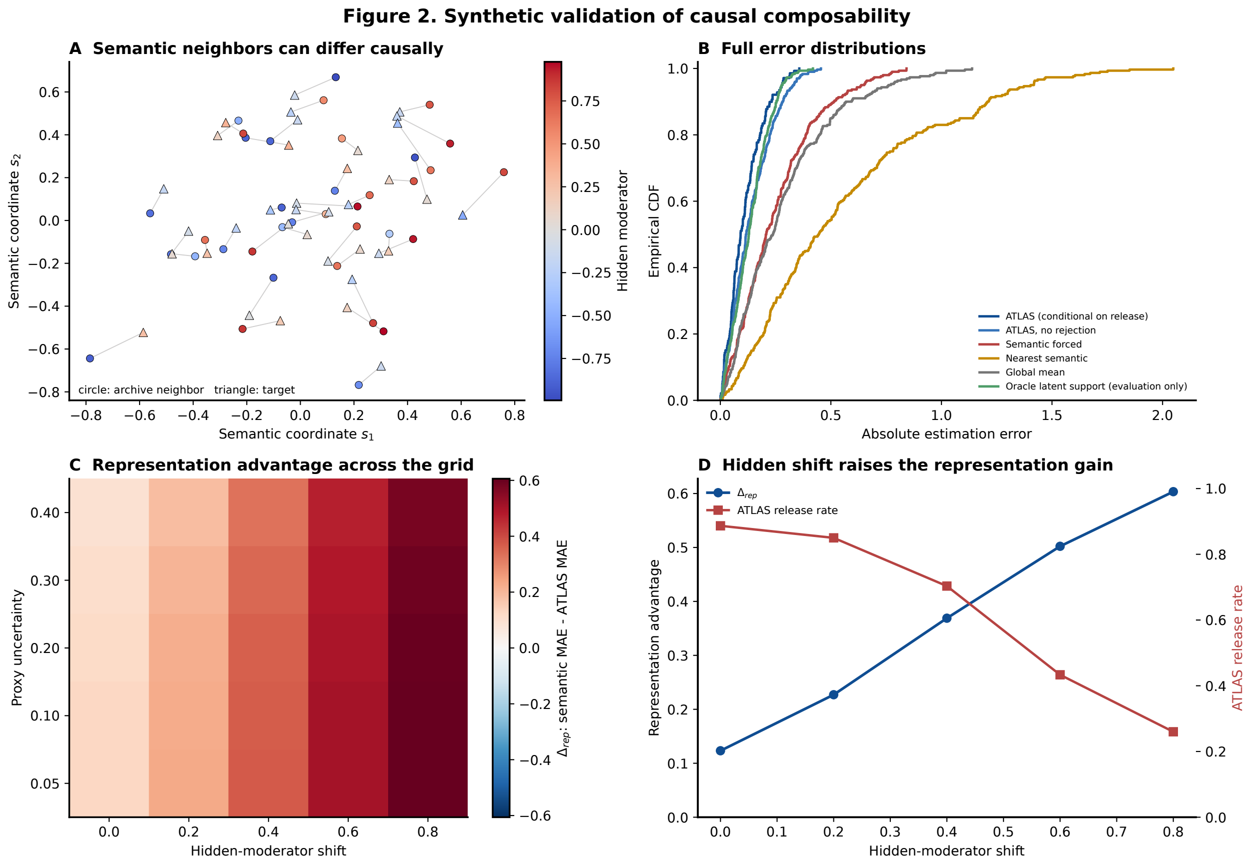

Figure 2: Synthetic validation of causal composability. **(a)** Semantic neighbors can differ in the hidden moderator. **(b)** Full absolute-error distributions for the six shared-target methods; the released ATLAS curve is conditional on the 139 released targets. **(c)** Representation advantage $\Delta_{\text{rep}} = \text{MAE}_{\text{semantic}} - \text{MAE}_{\text{ATLAS,no-rejection}}$ over hidden shift and proxy uncertainty. **(d)** At proxy uncertainty 0.10, increasing hidden-moderator shift increases the representation advantage while reducing the ATLAS release rate.

Figure 2 explains why the mean comparison is not merely a semantic-neighbor result. Targets can be close in (s_1, s_2) while differing materially in h, so methods that use only semantic coordinates exhibit heavier error tails than the design-enriched representation. The behavior is consistent with the non-identifiability phenomenon in Theorem 5.3; it is an empirical illustration rather than a proof of that theorem.

The representation sensitivity grid isolates this mechanism by disabling rejection in the ATLAS-versus-semantic comparison. At proxy uncertainty 0.10, increasing the hidden shift from 0 to 0.8 increases Δ_{rep} from 0.123 to 0.603. At the same time, the ATLAS release rate falls from 0.887 to 0.260. Thus a representation can improve prediction while the certificate correctly determines that fewer targets have enough support for release.

A separate target-level certificate diagnostic is reported in Appendix Figure 6; its certificate radius has Spearman correlation 0.272 with realized absolute error across the 300 common targets. No realized error exceeds the reported certificate radius in this diagnostic set, which is descriptive rather than a pointwise calibration guarantee.

Figure 3. Selective prediction and honest uncertainty

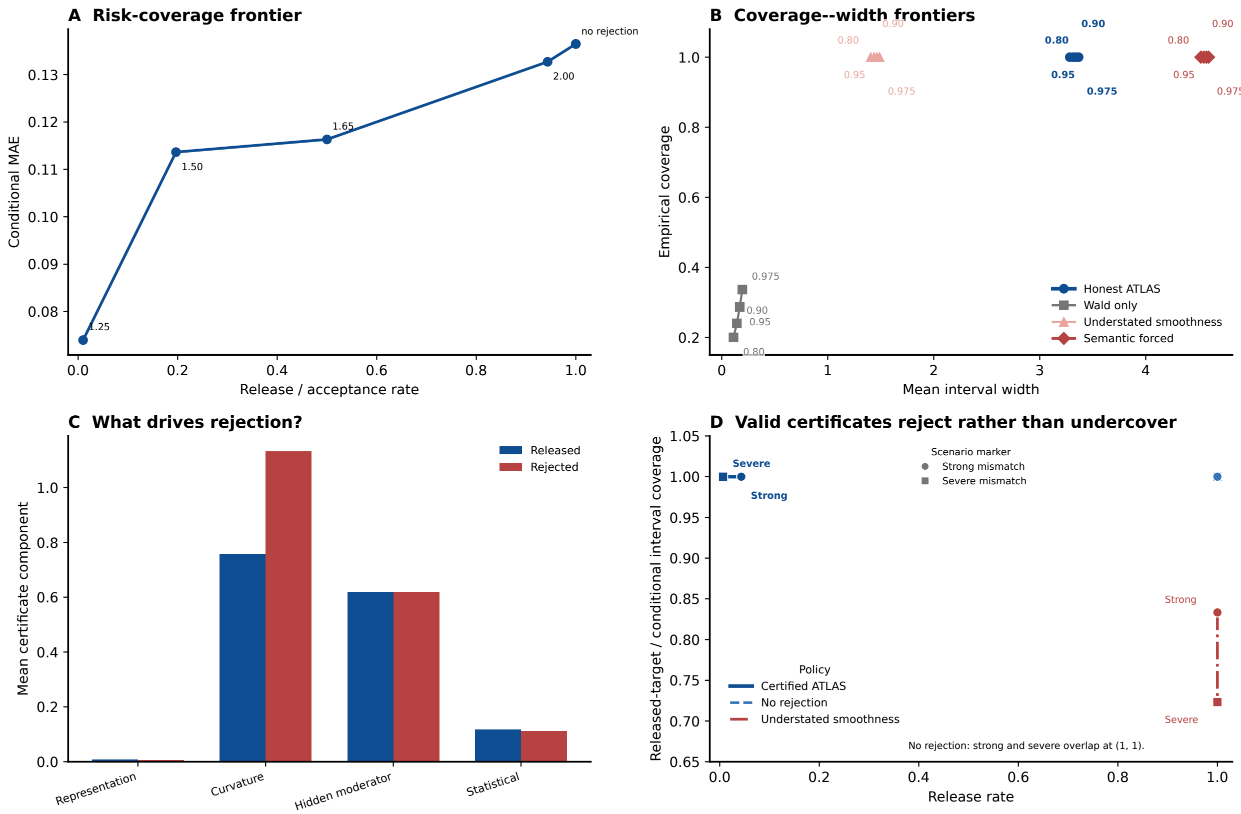

Figure 3: Selective prediction and honest uncertainty. **(a)** Risk–coverage frontier on a fixed target set. **(b)** Empirical coverage versus mean interval width across confidence levels and interval policies. **(c)** Mean certificate-component decomposition for released and rejected targets, showing that curvature/support uncertainty is the main discriminator in the nominal DGP. **(d)** Released-target coverage boundary under strong and severe mismatch when smoothness is deliberately understated. The honest interval is conservative because it retains approximation and moderator uncertainty; Wald-only intervals omit those components.

6.3 Selective Prediction and Honest Uncertainty

Selective prediction must be evaluated as a risk–coverage tradeoff rather than as a single conditional error. Holding the same 300 nominal targets fixed and changing only the scientific tolerance gives release rates 0.010, 0.500, and 0.943 at thresholds 1.25, 1.65, and 2.00, with conditional MAEs 0.074, 0.116, and 0.133; the no-rejection endpoint has MAE 0.136 (Figure 3a). Relaxing the threshold therefore exposes an explicit frontier instead of hiding the cost of abstention. Across nominal confidence levels 0.80, 0.90, 0.95, and 0.975, the honest intervals have empirical coverage 1.000 while mean width increases from 3.283 to 3.367. The coverage–width frontier in Figure 3b therefore makes the cost of honest uncertainty explicit. The certificate decomposition in Figure 3c clarifies why targets are rejected in the nominal DGP: the curvature component rises from 0.759 among released targets to 1.131 among rejected targets, whereas the hidden-moderator and statistical components remain nearly unchanged. Thus rejection here is driven primarily by geometric/curvature uncertainty rather than sampling noise. Wald-only intervals have coverage only 0.200, 0.240, 0.287, and 0.337, with widths below 0.20, showing why sampling uncertainty alone is insufficient for archive transport.

The negative-control experiments make the assumption boundary explicit. Figure 3d shows that, under strong and severe semantic mismatch, deliberately understating the smoothness bounds lowers coverage to 0.833 and 0.723, respectively, while the certified procedure continues to reject almost all unsupported targets and retains conservative coverage on the released branch. The conclusion is conditional: the guarantees depend on valid certificate components, and a numerically narrow but misspecified certificate can be overconfident.

6.4 From Rejection to Partial Identification

Rejection is not a terminal failure. We move the target toward a fixed in-domain anchor and, after a point prediction is rejected, return the Theorem 5.4 partial-identification interval. As support deteriorates, the rejection rate increases from 0.517 to 0.990, while the average PI diameter on rejected targets increases from 3.646 to 7.076 (Table 2). Figure 4a visualizes the same monotone widening of the identified set. All rejected-branch intervals contain the simulated truth in the prespecified scenarios, and all intersections are nonempty. The width increase is therefore the intended expression of loss of identification rather than a numerical failure. In short,

$$\text{support worsens} \;\Rightarrow\; \text{release decreases} \;\Rightarrow\; \text{the identified set widens.}$$

Table 2: Support deterioration, rejection, and partial identification (PI) under the shared-target support-stress protocol. All PI quantities are evaluated on rejected synthetic targets.

Scenario	Rejection	Nonempty PI	PI coverage	PI diameter
Nominal support	0.517	1.000	1.000	3.646
Moderate mismatch	0.777	1.000	1.000	4.095
Strong mismatch	0.973	1.000	1.000	5.674
Severe mismatch	0.990	1.000	1.000	7.076

6.5 Bridge Experiment Design

Once composition is rejected, the next question is not how to extrapolate harder, but which experiment should be run next. The bridge planner evaluates the conditional marginal expected reduction in the current PI diameter, selects a candidate, observes its noisy effect, and recomputes the remaining values. We compare causal-support greedy selection with semantic-only greedy and uniform random selection under a budget of four.

In the severe-mismatch formal experiment, causal greedy reduces the mean diameter from 6.942 to 1.877, a 72.96% reduction. Semantic greedy and random finish at 2.331 and 2.317, respectively, so their performance is substantially closer to each other than to the causal-support policy. The focused path in Figure 4b shows the same ordering while exposing the budget trajectory. Figure 4c provides an evaluation-only mechanism check: causal-support bridge selection reduces the target distance to the expanded true-mechanism hull, although those latent coordinates are never available to the planner.

For the small candidate library where ex-post enumeration is feasible, greedy attains 0.9957, 0.9776, and 0.9857 of the exhaustive bridge value at budgets 1, 2, and 3; Figure 4d reports these value ratios. Because the exhaustive comparator uses realized bridge outcomes, it is not deployable and does not estimate a weak-submodularity parameter or prove the coefficient in Theorem 5.6.

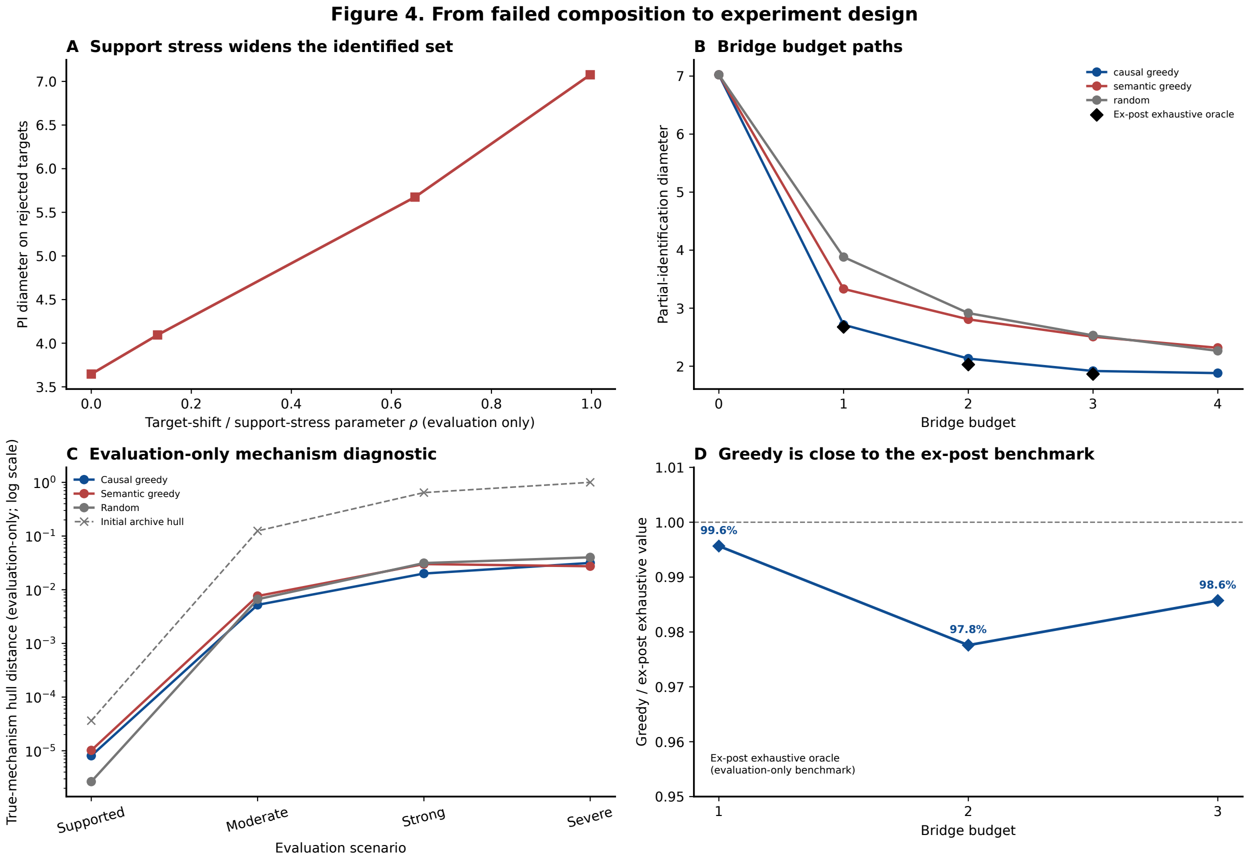

Figure 4: From failed composition to experiment design. **(a)** Partial-identification diameter on rejected targets as the evaluation-only support-stress parameter increases. **(b)** Sequential PI-diameter trajectories as bridge budget increases under severe mismatch; the focused visualization uses 90 paths, whereas formal policy claims use 300 repetitions. Black diamonds denote the evaluation-only ex-post exhaustive comparator where available. **(c)** Evaluation-only distance from the target mechanism to the expanded true-mechanism hull across support scenarios; latent mechanisms are never available to the planner. **(d)** Greedy bridge value relative to the ex-post exhaustive benchmark for budgets 1–3.

6.6 Real-Data NSW Archive

The real-data evaluation uses the Dehejia–Wahba NSW job-training data (LaLonde, 1986; Dehejia and Wahba, 1999). We use the randomized NSW job-training data as a reconstruction stress test rather than as a setting with known subgroup-level causal ground truth. Local archive objects are constructed from covariate neighborhoods with minimum treated and control support; held-out objects are reconstructed from the remaining archive. The held-out local contrast is a noisy reference quantity, not the true conditional treatment effect.

Causal ATLAS has all-target MAE 0.862 and released-target MAE 0.864, median absolute error 0.699, sign accuracy 0.854, held-out reference inclusion 0.974, width 5.630, and rejection rate 0.232. Semantic forced and global mean are narrower but include the held-out reference only 0.605 and 0.446 of the time, whereas nearest semantic reaches 0.992 inclusion with width 8.311. All-target point metrics are computed on the same 1,680 raw reconstructions, so Causal ATLAS and its no-rejection counterpart have identical all-target MAE, median absolute error, and sign

Table 3: NSW held-out local-contrast reconstruction. All-target MAE uses all 1,680 raw reconstructions; Released MAE uses released targets only. Inclusion refers to a noisy held-out local contrast, not a latent subgroup effect; monetary quantities are thousands of U.S. dollars.

Method	All-target MAE	Released MAE	Median AE	Sign	Inclusion	Width	Rejection
Causal ATLAS	0.862	0.864	0.699	0.854	0.974	5.630	0.232
ATLAS, no rejection	0.862	0.862	0.699	0.854	0.970	5.275	0.000
Semantic forced	1.169	1.169	0.937	0.754	0.605	2.854	0.000
Nearest semantic	1.242	1.242	1.031	0.780	0.992	8.311	0.000
Global mean	1.679	1.679	1.349	0.784	0.446	2.475	0.000

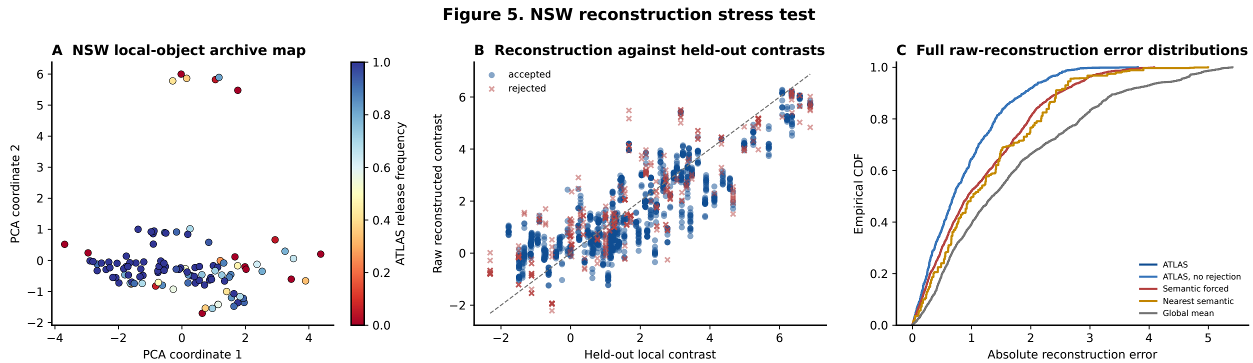

Figure 5: NSW reconstruction stress test. **(a)** PCA map of the 112 local archive objects, colored by ATLAS release frequency. **(b)** Raw reconstruction against noisy held-out local contrasts, with released and rejected ATLAS targets distinguished. **(c)** Full empirical distributions of absolute raw-reconstruction error for ATLAS and the baseline methods. Inclusion refers to a noisy held-out reference, not coverage of a latent subgroup causal effect.

accuracy. The released ATLAS branch has MAE 0.864, essentially the same as the all-target MAE 0.862, indicating that rejection in this NSW stress test primarily affects certification and reporting rather than selectively lowering reconstruction error. Figure 5c shows the complete absolute-error distributions, confirming that the aggregate MAE ordering is not driven by a small number of extreme reconstructions. These results reproduce the reconstruction–certification tradeoff in a heterogeneous archive, not a real-world causal generalization claim.

The archive map and target-level diagnostics show that reconstruction becomes harder in sparsely supported regions. Local objects overlap, so their estimated effects are statistically dependent; the current experiment is descriptive and does not implement a full shared-unit covariance correction. The results therefore support the practical behavior of the framework while leaving external validity, alternative neighborhood definitions, and nuisance-estimation uncertainty for future work.

Across the synthetic experiments, Causal ATLAS improves composition when causal support is informative, exposes a transparent risk–coverage frontier, and protects interval validity when its declared uncertainty bounds are correct. When support is insufficient, it transitions to partial identification instead of forced extrapolation; bridge experiments then shrink the resulting diameter. NSW is consistent with these mechanisms as a reconstruction stress test, while supplying no causal ground truth for subgroup effects.

7 Discussion

This paper draws a boundary around what an experimental archive can and cannot do. If the representation is causally sufficient, the design profiles are compatible, and the target lies in local support, then the archive can produce a debiased, certified transported effect. If a relevant moderator is omitted, semantic proximity is not enough. The correct output is a partial-identification region, a sensitivity statement, and a bridge-design recommendation.

The limitations are best understood as new research questions. First, the mechanism representation is treated as supplied by diagnostics, metadata, and analyst encoding. Learning such representations from multimodal experiment descriptions requires separate identifiability conditions; this is not a mere representation-learning problem because omitted moderators induce non-identification. Second, local smoothness is a scope condition rather than a universal law. Negative controls, placebo outcomes, moderator curves, and sensitivity analysis should be used to decide where smoothness is credible. Third, bridge value is computed over a finite candidate library. Designing the library itself is an experimental-design problem: one must choose which populations, treatment intensities, outcomes, and control conditions could most efficiently close the support gap.

The broader implication is that scientific discovery engines should be allowed to say no. A system that always returns a point estimate hides the distinction between evidence reuse and extrapolation. A causal atlas makes that distinction explicit. It can say: this effect transports; this effect is only partially identified; and this is the bridge experiment that would change the answer. In the Causal ATLAS roadmap, this paper supplies the first module: archive-to-discovery evidence reuse with honest refusal. Later modules can build on it for LLM-generated treatments, regret-calibrated assignment, causal routing, dynamic networks, strategic AI systems, geometric identifiability, and discovery engines.

A Appendix: Notation and Proofs

A.1 Notation Table

Table 4: Core notation used in the proofs.

Symbol	Meaning
e_i	Archived experiment object
$e_\star$	Target experiment object
$W_{i\ell}$	Unit record $(X_{i\ell}, A_{i\ell}, Y_{i\ell})$ in experiment i
n_i	Sample size of experiment i
$m_i, m_\star$	Latent causal mechanism vectors in $\mathcal{M} \subset \mathbb{R}^p$
$r_i, r_\star$	Observed atlas representations
$\tau_i, \tau_\star$	True causal effects $\mu(m_i), \mu(m_\star)$
$\widehat{\tau}_i$	Debiased archive effect estimate
$\mathcal{U}_i = (n_i, \widehat{v}_i, \mathfrak{b}_i)$	Experiment-level uncertainty certificate
μ	Effect surface on the mechanism space
$\kappa(C, \star)$	Design-compatibility predicate for candidate set C and target
Δ_C	Simplex over candidate experiments C
α	Composition weights
$\bar{m}_\alpha$	Weighted mechanism $\sum_{j \in C} \alpha_j m_j$
$\rho_\star(\alpha)$	Causal support residual $\Vert m_\star - \bar{m}_\alpha \Vert$
$R_{\mathrm{hid}}(\alpha)$	Hidden-moderator sensitivity residual
$\mathfrak{b}_i$	Second-order nuisance-bias bound for experiment i
$\Theta_A(e_\star)$	Archive-induced partial-identification region
$\mathrm{VoI}_A(b)$	Bridge value of information for candidate bridge b

The appendix follows the logical order of the main results. It first gives the standard conditional-unbiasedness and nuisance-error calculation used to construct $\mathfrak{b}_i$, followed by the proof of the auxiliary variance envelope in Lemma 5.1. It then verifies the interval construction in Lemma 5.2 and proves the finite-sample composition certificate in Theorem 5.1. The classical affine boundary case needs no separate proof. The normalized triangular-array and confidence-interval arguments are presented together as the proof of Theorem 5.2. The remaining subsections prove the semantic boundary, partial-identification, minimax, and bridge-design results in their order of appearance.

A.2 AIPW Orthogonality and Nuisance-Remainder Calculation

Proof strategy. We condition on the nuisance-training folds first. The proof then evaluates one score at a time, derives its conditional mean, and controls the second-order remainder used in the composition radius.

The score used throughout this proof is the AIPW score in (4.1), while its cross-fitted average is the estimator in (4.2). We first establish the conditional-unbiasedness claim in (A.2), then derive the nuisance product remainder in (A.12).

Fix an experiment i and suppress the subscript. Let $I_1, \dots, I_K$ be its evaluation folds, let $\mathcal{T}_k$ be the data used to train the nuisance functions applied on I_k, and write $\widehat{\tau}_k$ for the fold average. Cross-fitting ensures that, conditional on $\mathcal{T}_k$, the fitted functions are fixed and the observations in I_k retain their evaluation law. Write $\mu_a(x) = \mathbb{E}(Y \mid A = a, X = x) = \mathbb{E}\{Y(a) \mid X = x\}$, where the last equality follows from consistency and ignorability. Conditional on the cross-fitting training folds, the nuisance functions entering the score are fixed functions of x. For arbitrary fixed functions $\widehat{\mu}_0, \widehat{\mu}_1$ and the true propensity π, define

$$\phi(W; \widehat{\mu}_0, \widehat{\mu}_1, \pi) = \widehat{\mu}_1(X) - \widehat{\mu}_0(X) + \frac{A\{Y - \widehat{\mu}_1(X)\}}{\pi(X)} - \frac{(1-A)\{Y - \widehat{\mu}_0(X)\}}{1-\pi(X)}. \tag{A.1}$$

The display in (A.1) defines the single-observation score. For known propensities, the precise claim needed for ordinary cross-fitting is the foldwise conditional identity

$$\mathbb{E}\{\widehat{\tau}_k \mid \mathcal{T}_k\} = \tau, \qquad \widehat{\tau}_k = \frac{1}{|I_k|} \sum_{\ell \in I_k} \phi(W_\ell; \widehat{\mu}_0^{(-k)}, \widehat{\mu}_1^{(-k)}, \pi). \tag{A.2}$$

Conditioning on $\mathcal{T}_k$ permits the fold-average reduction in (A.3). We then condition once more on X to evaluate a generic score,

$$\begin{aligned} \mathbb{E}(\widehat{\tau}_k \mid \mathcal{T}_k) &= \frac{1}{|I_k|} \sum_{\ell \in I_k} \mathbb{E}\Big\{\phi(W_\ell; \widehat{\mu}_0^{(-k)}, \widehat{\mu}_1^{(-k)}, \pi) \mid \mathcal{T}_k\Big\} \\ &= \mathbb{E}\Big\{\phi(W; \widehat{\mu}_0^{(-k)}, \widehat{\mu}_1^{(-k)}, \pi) \mid \mathcal{T}_k\Big\}. \end{aligned} \tag{A.3}$$

$$\begin{aligned} &\mathbb{E}\Big\{\phi(W; \widehat{\mu}_0^{(-k)}, \widehat{\mu}_1^{(-k)}, \pi) \mid X, \mathcal{T}_k\Big\} \\ &= \widehat{\mu}_1^{(-k)}(X) - \widehat{\mu}_0^{(-k)}(X) + \mathbb{E}\left\{\frac{A\{Y - \widehat{\mu}_1^{(-k)}(X)\}}{\pi(X)} \middle| X, \mathcal{T}_k\right\} \\ &\quad - \mathbb{E}\left\{\frac{(1-A)\{Y - \widehat{\mu}_0^{(-k)}(X)\}}{1-\pi(X)} \middle| X, \mathcal{T}_k\right\}. \end{aligned} \tag{A.4}$$

The two inverse-propensity terms in (A.4) are evaluated separately: the treated-arm calculation is (A.5), and the control-arm calculation is (A.6).

$$\begin{aligned} &\mathbb{E}\left\{\frac{A\{Y - \widehat{\mu}_1^{(-k)}(X)\}}{\pi(X)} \middle| X, \mathcal{T}_k\right\} \\ &= \frac{\mathbb{P}(A=1 \mid X)}{\pi(X)} \mathbb{E}\{Y - \widehat{\mu}_1^{(-k)}(X) \mid A=1, X, \mathcal{T}_k\} \\ &= \mu_1(X) - \widehat{\mu}_1^{(-k)}(X). \end{aligned} \tag{A.5}$$

$$\begin{aligned} &\mathbb{E}\left\{\frac{(1-A)\{Y - \widehat{\mu}_0^{(-k)}(X)\}}{1-\pi(X)} \middle| X, \mathcal{T}_k\right\} \\ &= \frac{\mathbb{P}(A=0 \mid X)}{1-\pi(X)} \mathbb{E}\{Y - \widehat{\mu}_0^{(-k)}(X) \mid A=0, X, \mathcal{T}_k\} \\ &= \mu_0(X) - \widehat{\mu}_0^{(-k)}(X). \end{aligned} \tag{A.6}$$

Substitution of (A.5) and (A.6) gives the cancellation displayed in (A.7); the resulting conditional causal contrast is recorded in (A.8).

$$\begin{aligned} &\mathbb{E}\Big\{\phi(W; \widehat{\mu}_0^{(-k)}, \widehat{\mu}_1^{(-k)}, \pi) \mid X, \mathcal{T}_k\Big\} \\ &= \widehat{\mu}_1^{(-k)}(X) - \widehat{\mu}_0^{(-k)}(X) + \mu_1(X) - \widehat{\mu}_1^{(-k)}(X) - \mu_0(X) + \widehat{\mu}_0^{(-k)}(X) \\ &= \mu_1(X) - \mu_0(X). \end{aligned} \tag{A.7}$$

$$
\begin{aligned}
\mathbb{E}(\widehat{\tau}_k \mid \mathcal{T}_k) &= \mathbb{E}_X\{\mu_1(X) - \mu_0(X)\} = \mathbb{E}\{Y(1) - Y(0)\} = \tau, \\
\mathbb{E}(\widehat{\tau}) &= \sum_{k=1}^{K} \frac{|I_k|}{n} \mathbb{E}(\widehat{\tau}_k) = \tau.
\end{aligned} \tag{A.8}
$$

Averaging the identity in (A.8) over X gives $\mathbb{E}\{\phi(W; \widehat{\mu}_0, \widehat{\mu}_1, \pi)\} = \tau$. Taking expectations over each training sample and then forming the fold-size-weighted average gives the second line of (A.8), proving the first conclusion. A stronger identity conditional on every fitted nuisance requires external training and is not needed for ordinary cross-fitting. The Horvitz–Thompson estimator is obtained by taking $\widehat{\mu}_0 = \widehat{\mu}_1 = 0$, so its unbiasedness follows as a special case.

When the propensity is estimated, let $\widehat{\pi}$ be its clipped estimate. Conditional on X and the nuisance training folds,

$$
\begin{aligned}
G_k(x) &:= \mathbb{E}\Big\{\phi(W; \widehat{\mu}_0^{(-k)}, \widehat{\mu}_1^{(-k)}, \widehat{\pi}^{(-k)}) \mid X = x, \mathcal{T}_k\Big\} \\
&\qquad - \{\mu_1(x) - \mu_0(x)\}, \\
\mathbb{E}(\widehat{\tau}_k - \tau \mid \mathcal{T}_k) &= \mathbb{E}_X\{G_k(X) \mid \mathcal{T}_k\}.
\end{aligned} \tag{A.9}
$$

Equation (A.9) defines the pointwise conditional score error G_k and then averages it over the evaluation covariate distribution. Expanding that error and collecting the two nuisance discrepancies yields the product form in (A.10).

$$
\begin{aligned}
G_k(X) &= \widehat{\mu}_1^{(-k)}(X) - \widehat{\mu}_0^{(-k)}(X) + \frac{\pi(X)}{\widehat{\pi}^{(-k)}(X)}\{\mu_1(X) - \widehat{\mu}_1^{(-k)}(X)\} \\
&\qquad - \frac{1 - \pi(X)}{1 - \widehat{\pi}^{(-k)}(X)}\{\mu_0(X) - \widehat{\mu}_0^{(-k)}(X)\} - \mu_1(X) + \mu_0(X) \\
&= \{\widehat{\pi}^{(-k)}(X) - \pi(X)\}\left\{\frac{\widehat{\mu}_1^{(-k)}(X) - \mu_1(X)}{\widehat{\pi}^{(-k)}(X)} + \frac{\widehat{\mu}_0^{(-k)}(X) - \mu_0(X)}{1 - \widehat{\pi}^{(-k)}(X)}\right\}.
\end{aligned} \tag{A.10}
$$

Using the clipping condition $\widehat{\pi}^{(-k)} \in [\pi_0, 1 - \pi_0]$, the triangle inequality, and Cauchy–Schwarz gives (A.11).

$$
\begin{aligned}
&|\mathbb{E}(\widehat{\tau}_k - \tau \mid \mathcal{T}_k)| = |\mathbb{E}_X\{G_k(X) \mid \mathcal{T}_k\}| \\
&\le \frac{1}{\pi_0}\mathbb{E}_X\left[\left|\widehat{\pi}^{(-k)}(X) - \pi(X)\right|\left\{\left|\widehat{\mu}_1^{(-k)}(X) - \mu_1(X)\right| + \left|\widehat{\mu}_0^{(-k)}(X) - \mu_0(X)\right|\right\}\Big|\mathcal{T}_k\right] \\
&\le \frac{1}{\pi_0}\left\|\widehat{\pi}^{(-k)} - \pi\right\|_2\left\{\left\|\widehat{\mu}_1^{(-k)} - \mu_1\right\|_2 + \left\|\widehat{\mu}_0^{(-k)} - \mu_0\right\|_2\right\}.
\end{aligned} \tag{A.11}
$$

Here $\|U\|_2 = \mathbb{E}(U^2)^{1/2}$, with the expectation taken under the evaluation distribution. Averaging the foldwise Cauchy–Schwarz bounds over the training samples yields the archive-level second-order remainder in (A.12).

$$
\begin{aligned}
|\mathbb{E}(\widehat{\tau} - \tau)| &\le \sum_{k=1}^{K} \frac{|I_k|}{n}\left|\mathbb{E}(\widehat{\tau}_k - \tau)\right| \\
&\le \sum_{k=1}^{K} \frac{|I_k|}{n}\mathbb{E}\left[\frac{1}{\pi_0}\left\|\widehat{\pi}^{(-k)} - \pi\right\|_2\left\{\left\|\widehat{\mu}_1^{(-k)} - \mu_1\right\|_2 + \left\|\widehat{\mu}_0^{(-k)} - \mu_0\right\|_2\right\}\right] =: \mathfrak{b}.
\end{aligned} \tag{A.12}
$$

This is the product-error certificate in (4.3). Together with the known-propensity calculation above, it supplies the standard experiment-level facts used by the composition and asymptotic arguments.

A.3 Proof of Lemma 5.1

Proof strategy. Independent external nuisance training makes the evaluation scores conditionally independent and identically distributed. We combine the resulting variance reduction with a clipped-score envelope.

Under that common external-training setup, conditional independence of evaluation units gives

$$\begin{aligned}\operatorname{Var}(\widehat{\tau} \mid \widehat{\eta}) &= \operatorname{Var}\left\{\frac{1}{n}\sum_{\ell=1}^{n}\phi(W_\ell;\widehat{\eta})\middle|\widehat{\eta}\right\} \\ &= \frac{1}{n^2}\sum_{\ell=1}^{n}\operatorname{Var}\{\phi(W_\ell;\widehat{\eta}) \mid \widehat{\eta}\} = \frac{1}{n}\operatorname{Var}\{\phi(W;\widehat{\eta}) \mid \widehat{\eta}\},\end{aligned} \tag{A.13}$$

Under clipping, $|\widehat{\mu}_1(X) - \widehat{\mu}_0(X)| \leq 2M$. On the event $A = 1$, $|Y - \widehat{\mu}_1(X)| \leq 2M$, and on the event $A = 0$, $|Y - \widehat{\mu}_0(X)| \leq 2M$. Since only one inverse-propensity term is active and the denominator is at least π_0,

$$|\phi(W;\widehat{\eta})| \leq 2M + 2M/\pi_0 = C_M. \tag{A.14}$$

Thus $\operatorname{Var}\{\phi(W;\widehat{\eta}) \mid \widehat{\eta}\} \leq C_M^2$. For independent archive experiments, the variance of a weighted sum is the weighted sum of variances, giving the final bound.

A.4 Proof of Lemma 5.2

Proof strategy. We verify the claim in two steps: each candidate certificate is an interval, and intersections of intervals retain the interval property.

The candidate interval used in the definition of the estimated partial-identification region is exactly $I(\alpha)$ in (A.15). Thus the proof below checks the two ingredients needed for Lemma 5.2: closedness of each certificate and preservation of the interval property under intersection.

For a fixed $\alpha \in \mathcal{W}_A(e_\star)$, define

$$I(\alpha) = \left[\sum_{j\in C}\alpha_j\widehat{\tau}_j - B(\alpha), \sum_{j\in C}\alpha_j\widehat{\tau}_j + B(\alpha)\right]. \tag{A.15}$$

Each $I(\alpha)$ in (A.15) is a closed interval in $\mathbb{R}$. The intersection of closed subsets of $\mathbb{R}$ is closed. If the intersection is nonempty and $x < y < z$ with x, z in the intersection, then $x, z \in I(\alpha)$ for every α. Since every $I(\alpha)$ is an interval, $y \in I(\alpha)$ for every α, hence y lies in the intersection. Therefore the intersection is a closed interval, which is the well-posedness conclusion of Lemma 5.2. If the intersection is empty, no value of $\tau_\star$ can satisfy all asserted certificates simultaneously, so the archive estimates and certificates are mutually inconsistent at the chosen level.

A.5 Proof of Theorem 5.1

Proof strategy. The proof separates deterministic approximation error from statistical error. We first control the mechanism geometry and curvature, then add the hidden-moderator term and the stochastic concentration term.

The target is the deterministic radius in (5.1), whose support, curvature, hidden-moderator, nuisance, and stochastic components are assembled below. The support contribution is controlled by the line-integral calculation in (A.16); the curvature contribution is derived in (A.17)–(A.20); and the resulting causal transport inequality is summarized in (A.22).

Fix C and $\alpha \in \Delta_C$. Write $\bar{m}_\alpha = \sum_{j \in C} \alpha_j m_j$. Define the scalar path $g(t) = \mu\{\bar{m}_\alpha + t(m_\star - \bar{m}_\alpha)\}$. The fundamental theorem of calculus and Cauchy–Schwarz give

$$\begin{aligned} g(t) &:= \mu\{\bar{m}_\alpha + t(m_\star - \bar{m}_\alpha)\}, \qquad g(1) = \mu(m_\star), \quad g(0) = \mu(\bar{m}_\alpha), \\ |\mu(m_\star) - \mu(\bar{m}_\alpha)| &= |g(1) - g(0)| = \left| \int_0^1 g'(t)\, dt \right| \\ &= \left| \int_0^1 \langle \nabla \mu\{\bar{m}_\alpha + t(m_\star - \bar{m}_\alpha)\}, m_\star - \bar{m}_\alpha \rangle \, dt \right| \\ &\leq \int_0^1 \|\nabla \mu\{\bar{m}_\alpha + t(m_\star - \bar{m}_\alpha)\}\| \, \|m_\star - \bar{m}_\alpha\| \, dt \\ &\leq L \|m_\star - \bar{m}_\alpha\| = L \rho_\star(\alpha). \end{aligned} \tag{A.16}$$

Equation (A.16) is the first-order support bound: it turns the mechanism distance $\rho_\star(\alpha)$ into an effect-scale error. For each j, Taylor's theorem with integral remainder around $\bar{m}_\alpha$ gives

$$\mu(m_j) = \mu(\bar{m}_\alpha) + \langle \nabla \mu(\bar{m}_\alpha), m_j - \bar{m}_\alpha \rangle + R_j, \tag{A.17}$$

$$R_j = \int_0^1 (\nabla \mu(\bar{m}_\alpha + t(m_j - \bar{m}_\alpha)) - \nabla \mu(\bar{m}_\alpha))^T (m_j - \bar{m}_\alpha) dt \tag{A.18}$$

where the H-Lipschitz gradient condition implies

$$\begin{aligned} |R_j| &\leq \int_0^1 \|\nabla \mu\{\bar{m}_\alpha + t(m_j - \bar{m}_\alpha)\} - \nabla \mu(\bar{m}_\alpha)\| \, \|m_j - \bar{m}_\alpha\| \, dt \\ &\leq H \int_0^1 t \|m_j - \bar{m}_\alpha\|^2 \, dt = \frac{H}{2} \|m_j - \bar{m}_\alpha\|^2 . \end{aligned} \tag{A.19}$$

The Taylor expansion in (A.17) defines the remainder R_j, whose integral representation is (A.18). The Lipschitz-gradient assumption bounds that remainder as shown in (A.19). Multiplying by α_j and summing, the first-order terms cancel because of the simplex identity in (A.20).

$$\sum_{j \in C} \alpha_j (m_j - \bar{m}_\alpha) = 0. \tag{A.20}$$

Therefore

$$\left| \sum_j \alpha_j \mu(m_j) - \mu(\bar{m}_\alpha) \right| = \left| \sum_j \alpha_j R_j \right| \leq \frac{H}{2} \sum_j \alpha_j \|m_j - \bar{m}_\alpha\|^2 . \tag{A.21}$$

Thus (A.21) is the curvature contribution. Adding the hidden-moderator discrepancy from Assumption 3.5 gives

$$\begin{aligned} \left| \tau_\star - \sum_j \alpha_j \tau_j \right| &= \left| \mu(m_*) - \sum_j \alpha_j \mu(m_j) \right| \\ &= \left| \mu(m_*) - \mu(\bar{m}_\alpha) + \mu(\bar{m}_\alpha) - \sum_j \alpha_j \mu(m_j) \right| \\ &\leq |\mu(m_*) - \mu(\bar{m}_\alpha)| + \left| \mu(\bar{m}_\alpha) - \sum_j \alpha_j \mu(m_j) \right| \\ &\leq L \rho_\star(\alpha) + \frac{H}{2} \sum_j \alpha_j \|m_j - \bar{m}_\alpha\|^2 + R_{\text{hid}}(\alpha). \end{aligned} \tag{A.22}$$

The deterministic transport part of the proof is therefore exactly the right-hand side of (A.22). We now add statistical and nuisance-estimation error by decomposing

$$\sum_j \alpha_j \widehat{\tau}_j = \sum_j \alpha_j \tau_j + \sum_j \alpha_j \xi_j + \sum_j \alpha_j b_j. \tag{A.23}$$

The decomposition in (A.23) is converted into an absolute error bound by the triangle inequality in (A.24). Then

$$\begin{aligned}\left|\tau_* - \sum_j \alpha_j \widehat{\tau}_j\right| &= \left|\tau_* - \sum_j \alpha_j \tau_j - \sum_j \alpha_j \xi_j - \sum_j \alpha_j b_j\right| \\ &\le \left|\tau_* - \sum_j \alpha_j \tau_j\right| + \left|\sum_j \alpha_j \xi_j\right| + \left|\sum_j \alpha_j b_j\right| \end{aligned} \tag{A.24}$$

Let $\mathcal{G} = \sigma\{(D_j, C_j, \mathcal{A}_j) : j \in C\}$. By Assumption 3.4, $\sum_j \alpha_j \xi_j$ is conditionally sub-Gaussian given $\mathcal{G}$, with variance proxy $\sum_j \alpha_j^2 s_j^2$. Independence factorizes its moment generating function as shown in (A.25). Markov's inequality then gives the one-sided tail bound in (A.26); optimizing λ yields (A.27) and its lower-tail analogue (A.28). The two tails combine in (A.29).

$$\begin{aligned}\mathbb{E}\left\{e^{\lambda \sum_j \alpha_j \xi_j} \mid \mathcal{G}\right\} &= \prod_j \mathbb{E}\{e^{\lambda \alpha_j \xi_j} \mid \mathcal{G}\} \\ &\le \prod_j \exp\left(\frac{\lambda^2 \alpha_j^2 s_j^2}{2}\right) = \exp\left(\frac{\lambda^2}{2} \sum_j \alpha_j^2 s_j^2\right).\end{aligned} \tag{A.25}$$

By Markov's inequality,

$$\begin{aligned}\mathbb{P}\left(\sum_j \alpha_j \xi_j \ge t \mid \mathcal{G}\right) &= \mathbb{P}\left(e^{\lambda \sum_j \alpha_j \xi_j} \ge e^{\lambda t} \mid \mathcal{G}\right) \\ &\le e^{-\lambda t} \mathbb{E}\left\{e^{\lambda \sum_j \alpha_j \xi_j} \mid \mathcal{G}\right\} \\ &\le \exp\left\{-\lambda t + \frac{\lambda^2}{2} \sum_j \alpha_j^2 s_j^2\right\}, \qquad \lambda > 0.\end{aligned} \tag{A.26}$$

Let $\lambda = \frac{t}{\sum_j \alpha_j^2 s_j^2}$, and

$$\mathbb{P}\left(\sum_j \alpha_j \xi_j \ge t \mid \mathcal{G}\right) \le \exp\left\{-\frac{t^2}{2\sum_j \alpha_j^2 s_j^2}\right\}, \qquad \lambda = \frac{t}{\sum_j \alpha_j^2 s_j^2}. \tag{A.27}$$

Similarly,we have

$$\mathbb{P}\left(\sum_j \alpha_j \xi_j \le -t \mid \mathcal{G}\right) \le \exp\left\{-\frac{t^2}{2\sum_j \alpha_j^2 s_j^2}\right\}. \tag{A.28}$$

so

$$\mathbb{P}\left(\left|\sum_j \alpha_j \xi_j\right| \ge t \mid \mathcal{G}\right) \le 2\exp\left\{-\frac{t^2}{2\sum_j \alpha_j^2 s_j^2}\right\}. \tag{A.29}$$

The choice $t = \sqrt{2\log(2/\zeta)\sum_j \alpha_j^2 s_j^2}$ converts (A.29) into the conditional high-probability event (A.30).

$$\mathbb{P}\left(\left|\sum_j \alpha_j \xi_j\right| \le \sqrt{2\log(2/\zeta)\sum_j \alpha_j^2 s_j^2}\ \middle|\ D_j, C_j, \mathcal{A}_j\right) \ge 1-\zeta. \tag{A.30}$$

The bias remainders satisfy $|b_j| \le \mathfrak{b}_j$. Combining the deterministic bound (A.22), the stochastic decomposition (A.24), and the event (A.30) gives, with probability at least $1-\zeta$,

$$\begin{aligned}\left|\tau_\star - \sum_j \alpha_j \widehat{\tau}_j\right| \le L\rho_\star(\alpha) + \frac{H}{2}\sum_j \alpha_j \left\|m_j - \bar{m}_\alpha\right\|^2 + R_{\text{hid}}(\alpha) \\ + \sum_j \alpha_j \mathfrak{b}_j + \sqrt{2\log(2/\zeta)\sum_j \alpha_j^2 s_j^2}.\end{aligned} \tag{A.31}$$

The resulting bound is displayed in (A.31). On this conditional event, the theorem's stated inequality is (5.2). Integrating over the conditioning variables proves the unconditional claim. The proxy-matrix-certified version follows because a sub-Gaussian vector with proxy matrix Σ has the one-dimensional proxy $\alpha^\top \Sigma \alpha$ in direction α.

A.6 Proof of Theorem 5.2

Proof strategy. We first rewrite the estimator as a normalized triangular-array sum plus a remainder. The Lindeberg–Feller theorem is then applied only to the leading sum, after which the remainder is handled by Slutsky's theorem.

The target of the calculation is the normalized limit stated in (5.3). Display (A.32) isolates the experiment-level influence term and nuisance remainder; the aggregation in (A.33)–(A.35) turns this into a triangular array. The next displays verify centering and unit variance before invoking the Lindeberg–Feller theorem.

Let $\psi_{j\ell} = \phi_j(W_{j\ell};\eta_j) - \tau_j$, $r_{j,n} = \frac{1}{n_j}\sum_{\ell=1}^{n_j}\{\phi_j(W_{j\ell};\widehat{\eta}_j^{(-\ell)}) - \phi_j(W_{j\ell};\eta_j)\}$

$$\begin{aligned}\widehat{\tau}_j - \tau_j &= \frac{1}{n_j}\sum_{\ell=1}^{n_j}\{\phi_j(W_{j\ell};\widehat{\eta}_j^{(-\ell)})\} - \tau_j \\ &= \frac{1}{n_j}\sum_{\ell=1}^{n_j}\{\phi_j(W_{j\ell};\widehat{\eta}_j^{(-\ell)}) - \phi_j(W_{j\ell};\eta_j) + \phi_j(W_{j\ell};\eta_j) - \tau_j\} \\ &= \frac{1}{n_j}\sum_{\ell=1}^{n_j}\psi_{j\ell} + r_{j,n},\end{aligned} \tag{A.32}$$

$$\begin{aligned}\therefore \frac{\widehat{\theta}_{\alpha_n} - \theta_{\alpha_n}}{V_{\alpha,n}^{1/2}} &= \frac{\sum_{j\in C_n}\alpha_{n,j}(\widehat{\tau}_j - \tau_j)}{V_{\alpha,n}^{1/2}} = \frac{1}{V_{\alpha,n}^{1/2}}\sum_{j\in C_n}\alpha_{n,j}\Big(\frac{1}{n_j}\sum_{\ell=1}^{n_j}\psi_{j\ell} + r_{j,n}\Big) \\ &= \sum_{j\in C_n}\sum_{\ell=1}^{n_j}\frac{\alpha_{n,j}}{n_j V_{\alpha,n}^{1/2}}\psi_{j\ell} + \frac{1}{V_{\alpha,n}^{1/2}}\sum_{j\in C_n}\alpha_{n,j} r_{j,n}\end{aligned} \tag{A.33}$$

where η_j denotes the true nuisance functions. The theorem explicitly assumes the aggregate asymptotic-linear remainder condition $\sum_j \alpha_{n,j} r_{j,n} = o_{\mathbb{P}}(V_{\alpha,n}^{1/2})$; cross-fitting and the product-rate calculation in Appendix A.2 are standard sufficient ingredients for verifying it. Therefore

$$\frac{1}{V_{\alpha,n}^{1/2}} \sum_{j \in C_n} \alpha_{n,j} r_{j,n} \xrightarrow{P} 0 \tag{A.34}$$

The probability statement in (A.34) makes the nuisance remainder negligible. Let $Z_{n,j\ell} = \frac{\alpha_{n,j}}{n_j V_{\alpha,n}^{1/2}} \psi_{j\ell}$ and $R_n = \sum_{j \in C_n} \alpha_{n,j} r_{j,n}$, so

$$\frac{\widehat{\theta}_{\alpha_n} - \theta_{\alpha_n}}{V_{\alpha,n}^{1/2}} = \sum_{j \in C_n} \sum_{\ell=1}^{n_j} Z_{n,j\ell} + \frac{R_n}{V_{\alpha,n}^{1/2}} \tag{A.35}$$

The normalized decomposition in (A.35) is the form to which the triangular-array CLT will be applied.

$$\because \mathbb{E}\{Z_{n,j\ell}\} = \frac{\alpha_{n,j}}{n_j V_{\alpha,n}^{1/2}} \mathbb{E}\{\psi_{j\ell}\} = 0, \therefore \mathbb{E}\{\sum_{j \in C_n} \sum_{\ell=1}^{n_j} Z_{n,j\ell}\} = 0 \tag{A.36}$$

Equation (A.36) verifies mean zero. The individual variances are computed in (A.37), summed in (A.38), and normalized to one in (A.39).

$$\operatorname{Var}(Z_{n,j\ell}) = \frac{\alpha_{n,j}^2}{n_j^2 V_{\alpha,n}} \operatorname{Var}(\psi_{j\ell}) = \frac{\alpha_{n,j}^2 v_j}{n_j^2 V_{\alpha,n}}. \tag{A.37}$$

$$\begin{aligned} \operatorname{Var}\left(\sum_{j \in C_n} \sum_{\ell=1}^{n_j} Z_{n,j\ell} \right) &= \sum_{j \in C_n} \sum_{\ell=1}^{n_j} \operatorname{Var}(Z_{n,j\ell}) \\ &= \frac{1}{V_{\alpha,n}} \sum_{j \in C_n} \frac{\alpha_{n,j}^2 v_j}{n_j}. \end{aligned} \tag{A.38}$$

$$V_{\alpha,n} = \sum_{j \in C_n} \frac{\alpha_{n,j}^2 v_j}{n_j} \quad \Longrightarrow \quad \operatorname{Var}\left(\sum_{j \in C_n} \sum_{\ell=1}^{n_j} Z_{n,j\ell} \right) = 1. \tag{A.39}$$

The unnormalized leading term has variance $V_{\alpha,n}$, while the normalized triangular array in (A.35) consists of independent, mean-zero summands with total variance one, as verified in (A.36)–(A.39). The stated Lindeberg condition therefore implies the Lindeberg–Feller central limit theorem, hence

$$\begin{aligned} \sum_{j \in C_n} \sum_{\ell=1}^{n_j} Z_{n,j\ell} &\rightsquigarrow \mathsf{N}(0,1) && \text{by Lindeberg–Feller,} \\ \frac{\widehat{\theta}_{\alpha_n} - \theta_{\alpha_n}}{V_{\alpha,n}^{1/2}} &= \sum_{j \in C_n} \sum_{\ell=1}^{n_j} Z_{n,j\ell} + o_{\mathbb{P}}(1) \rightsquigarrow \mathsf{N}(0,1) && \text{by Slutsky's theorem.} \end{aligned} \tag{A.40}$$

The convergence in (A.40) is the unstudentized conclusion. If $\widehat{V}_{\alpha,n}/V_{\alpha,n} \to_{\mathbb{P}} 1$, Slutsky's theorem gives the studentized result. Finally,

$$\widehat{\theta}_{\alpha_n} - \tau_\star = \widehat{\theta}_{\alpha_n} - \theta_{\alpha_n} + \theta_{\alpha_n} - \tau_\star. \tag{A.41}$$

The deterministic approximation bound implies $|\theta_{\alpha_n} - \tau_\star| \le D(\alpha_n)$. If $D(\alpha_n) = o(V_{\alpha,n}^{1/2})$, another application of Slutsky's theorem gives the final display. This decomposition is made explicit in (A.41) and (A.42).

$$\frac{\widehat{\theta}_{\alpha_n} - \tau_\star}{V_{\alpha,n}^{1/2}} = \frac{\widehat{\theta}_{\alpha_n} - \theta_{\alpha_n}}{V_{\alpha,n}^{1/2}} + \frac{\theta_{\alpha_n} - \tau_\star}{V_{\alpha,n}^{1/2}} \tag{A.42}$$

The approximation condition in (A.43) makes the second term in (A.42) converge to zero, so the centered limit is transferred from (A.40) to the target effect:

$$\because |\theta_{\alpha_n} - \tau_\star| \le D(\alpha_n), D(\alpha_n) = o(V_{\alpha,n}^{1/2}) \ \therefore \frac{\theta_{\alpha_n} - \tau_\star}{V_{\alpha,n}^{1/2}} \xrightarrow{P} 0 \tag{A.43}$$

$$\therefore \frac{\widehat{\theta}_{\alpha_n} - \tau_\star}{V_{\alpha,n}^{1/2}} \rightsquigarrow N(0,1) \tag{A.44}$$

Equation (A.44) is the target-scale asymptotic normality statement in Theorem 5.2; the preceding studentization argument gives its feasible-variance version.

Confidence-interval conclusion. The confidence statement in the same theorem follows by combining a statistical event for the centered estimator with a one-sided event ensuring that the estimated approximation radius is not too small.

The interval being justified is the honest interval in (5.4). The centered studentized statistic is controlled by (A.45); the deterministic triangle-inequality step is (A.46); and the two events controlling stochastic error and radius underestimation are combined in (A.47)– (A.50).

Along the triangular sequence of Theorem 5.2, suppress the index n to keep the notation readable. Then

$$\frac{\widehat{\theta}_\alpha - \theta_\alpha}{\widehat{V}_\alpha^{1/2}} \rightsquigarrow \mathsf{N}(0,1). \tag{A.45}$$

Consider the event EV_1 defined by $\tau_\star \in CI_{1-\zeta}(\alpha)$, which is equivalent to $\left|\widehat{\theta}_\alpha - \tau_\star\right| \le z_{1-\frac{\zeta}{2}} \widehat{V}_\alpha^{1/2} + \widehat{D}_\alpha$

$$\left|\widehat{\theta}_\alpha - \tau_\star\right| = \left|\widehat{\theta}_\alpha - \theta_\alpha + \theta_\alpha - \tau_\star\right| \le \left|\widehat{\theta}_\alpha - \theta_\alpha\right| + |\theta_\alpha - \tau_\star| \le \left|\widehat{\theta}_\alpha - \theta_\alpha\right| + D(\alpha) \tag{A.46}$$

$\forall\ 0 < \epsilon < z_{1-\frac{\zeta}{2}}$, consider the event EV_2 defined by $D(\alpha) - \widehat{D}(\alpha) \le \epsilon \widehat{V}_\alpha^{1/2}$ and event EV_3 defined by $\left|\widehat{\theta}_\alpha - \theta_\alpha\right| \le (z_{1-\frac{\zeta}{2}} - \epsilon)\widehat{V}_\alpha^{1/2}$,if both events occur,then

$$\left|\widehat{\theta}_\alpha - \tau_\star\right| \le (z_{1-\frac{\zeta}{2}} - \epsilon)\widehat{V}_\alpha^{1/2} + \widehat{D}(\alpha) + \epsilon \widehat{V}_\alpha^{1/2} = z_{1-\frac{\zeta}{2}} \widehat{V}_\alpha^{1/2} + \widehat{D}(\alpha) \tag{A.47}$$

Therefore, the event inclusion in

$$\{D(\alpha) - \widehat{D}(\alpha) \le \epsilon \widehat{V}_\alpha^{1/2}\} \cap \{\left|\widehat{\theta}_\alpha - \theta_\alpha\right| \le (z_{1-\frac{\zeta}{2}} - \epsilon)\widehat{V}_\alpha^{1/2}\} \subset \{\tau_\star \in CI_{1-\zeta}(\alpha)\} \tag{A.48}$$

Equation (A.48) is the formal implication from the two component events to interval coverage. Taking probabilities and using the studentized limit in (A.45) gives the lower-bound argument in (A.49)–(A.50).

$$\begin{aligned} \mathbb{P}\{\tau_\star \in \mathrm{CI}_{1-\zeta}(\alpha)\} &\ge \mathbb{P}(EV_2 \cap EV_3) \\ &\ge \mathbb{P}(EV_3) - \mathbb{P}(EV_2^c) \\ &= \mathbb{P}\left\{\left|\frac{\widehat{\theta}_\alpha - \theta_\alpha}{\widehat{V}_\alpha^{1/2}}\right| \le z_{1-\zeta/2} - \epsilon\right\} - \mathbb{P}\left\{\frac{D(\alpha) - \widehat{D}(\alpha)}{\widehat{V}_\alpha^{1/2}} > \epsilon\right\}. \end{aligned} \tag{A.49}$$

$$
\begin{aligned}
&\liminf_{n\to\infty} \mathbb{P}\{\tau_\star \in \mathrm{CI}_{1-\zeta}(\alpha)\} \geq 2\Phi(z_{1-\zeta/2} - \epsilon) - 1, \\
&\lim_{\epsilon \downarrow 0} \{2\Phi(z_{1-\zeta/2} - \epsilon) - 1\} = 1 - \zeta,
\end{aligned} \tag{A.50}
$$

The assumed positive-part condition, together with $\widehat{V}_\alpha / V_\alpha \to_{\mathbb{P}} 1$, implies $\mathbb{P}(EV_2^c) \to 0$ for every fixed $\epsilon > 0$. Equations (A.49)–(A.50) therefore give asymptotic coverage at least $1 - \zeta$. For the shorter Wald interval, assume additionally that $D(\alpha) = o(V_\alpha^{1/2})$. The detailed negligibility calculation is (A.51); the two convergences are displayed together in (A.52), and the Slutsky decomposition is (A.53). Consequently, (A.54) gives an asymptotic Wald-coverage statement.

$$
\begin{aligned}
&\left| \frac{\theta_\alpha - \tau_\star}{\widehat{V}_\alpha^{1/2}} \right| \leq \frac{D(\alpha)}{\widehat{V}_\alpha^{1/2}}, \\
D(\alpha) = o(V_\alpha^{1/2}), \quad \frac{\widehat{V}_\alpha}{V_\alpha} \xrightarrow{\mathbb{P}} 1 \quad &\Longrightarrow \quad \frac{\theta_\alpha - \tau_\star}{\widehat{V}_\alpha^{1/2}} \xrightarrow{\mathbb{P}} 0.
\end{aligned} \tag{A.51}
$$

$$
\frac{\widehat{\theta}_\alpha - \theta_\alpha}{\widehat{V}_\alpha^{1/2}} \rightsquigarrow \mathsf{N}(0,1), \qquad \frac{\theta_\alpha - \tau_\star}{\widehat{V}_\alpha^{1/2}} \xrightarrow{\mathbb{P}} 0. \tag{A.52}
$$

$$
\begin{aligned}
\frac{\widehat{\theta}_\alpha - \tau_\star}{\widehat{V}_\alpha^{1/2}} &= \frac{\widehat{\theta}_\alpha - \theta_\alpha}{\widehat{V}_\alpha^{1/2}} + \frac{\theta_\alpha - \tau_\star}{\widehat{V}_\alpha^{1/2}} \\
&\rightsquigarrow \mathsf{N}(0,1) \qquad \text{by Slutsky's theorem.}
\end{aligned} \tag{A.53}
$$

$$
\lim_{n\to\infty} \mathbb{P}\left\{ \left| \frac{\widehat{\theta}_\alpha - \tau_\star}{\widehat{V}_\alpha^{1/2}} \right| \leq z_{1-\zeta/2} \right\} = 1 - \zeta. \tag{A.54}
$$

A.7 Proof of Theorem 5.3

Proof strategy. The construction uses two experiments that are indistinguishable in the observed representation but differ in an omitted moderator. A smooth effect surface can therefore separate their causal effects.

This construction proves the impossibility claim in Theorem 5.3: the observed-distance premise is made explicit in (A.56), while the nonzero causal gap is computed in (A.57).

By assumption, the observed representation is many-to-one over an admissible moderator: there exist e^0, e^1 such that $x(e^0) = x(e^1)$ but $h(e^0) \neq h(e^1)$. Let $a = h(e^0)$, $b = h(e^1)$, and $a \neq b$. Define an effect surface

$$
\mu(m) = \lambda h(m), \tag{A.55}
$$

where $h(m)$ is the moderator coordinate and $\lambda > 0$ is chosen small enough that $\|\nabla \mu(m)\| \leq L$ over $\mathcal{M}$. This affine surface is defined in (A.55); hence it satisfies Assumption 3.3 with $H = 0$. The observed distance is

$$
\left\| x(e^0) - x(e^1) \right\| = 0 \leq \eta \tag{A.56}
$$

for every $\eta > 0$, while the causal effects differ by

$$
\left| \tau(e^1) - \tau(e^0) \right| = \left| \mu(m(e^1)) - \mu(m(e^0)) \right| = \lambda \left| b - a \right|. \tag{A.57}
$$

The distance equality (A.56) holds for every $\eta > 0$, whereas the effect difference is the positive constant in (A.57). Set $c = \lambda |b - a| > 0$. Any certificate based only on x-distance assigns the same distance information to the two experiments, yet their effects differ by c. Uniform causal-composability certification based only on observed semantic distance is therefore impossible over the stated model class.

A.8 Proof of Theorem 5.4

Proof strategy. We first establish simultaneous coverage by a finite union bound. We then construct two admissible effect surfaces that agree on the archive but disagree at an unsupported target.

The first part proves coverage of the reported set $\widehat{\Theta}_A(e_\star)$ from the single-composition certificate in (5.5); the second part constructs the non-identification pair underlying the diameter statement in Theorem 5.4.

For each $\alpha \in \mathcal{W}_A(e_\star)$, Theorem 5.1 implies the individual interval coverage statement

$$\mathbb{P}\left\{\tau_\star \in \left[\sum_j \alpha_j \widehat{\tau}_j - Rad(\alpha, \zeta_\alpha), \sum_j \alpha_j \widehat{\tau}_j + Rad(\alpha, \zeta_\alpha)\right]\right\} \geq 1 - \zeta_\alpha. \tag{A.58}$$

The individual interval in (A.58) is the building block for the intersection estimator. Consider the event defined by $\{\tau_* \in \widehat{\Theta}_A(e_*)\}$, whose set-theoretic form is

$$\because \quad \tau_* \in \widehat{\Theta}_A(e_*) \Longleftrightarrow \tau_* \in \bigcap_{\alpha \in \mathcal{W}_A(e_\star)} I_\alpha \tag{A.59}$$

The equivalence in (A.59) turns coverage of the intersection into a union-bound calculation. Applying the union bound gives (A.60), and the per-weight failure probability is bounded by (A.61).

$$\begin{aligned} \therefore \mathbb{P}\{\tau_* \in \widehat{\Theta}_A(e_*)\} &= \mathbb{P}\left\{\bigcap_{\alpha \in \mathcal{W}_A(e_\star)} \{\tau_* \in I_\alpha\}\right\} = 1 - \mathbb{P}\left\{\bigcup_{\alpha \in \mathcal{W}_A(e_\star)} \{\tau_* \notin I_\alpha\}\right\} \\ &\geq 1 - \sum_{\alpha \in \mathcal{W}_A(e_\star)} \mathbb{P}\{\tau_* \notin I_\alpha\} \end{aligned} \tag{A.60}$$

$$\because \mathbb{P}\{\tau_* \notin I_\alpha\} = \mathbb{P}\left\{\tau_\star \notin \left[\sum_j \alpha_j \widehat{\tau}_j - Rad(\alpha, \zeta_\alpha), \sum_j \alpha_j \widehat{\tau}_j + Rad(\alpha, \zeta_\alpha)\right]\right\} \leq \zeta_\alpha \tag{A.61}$$

$$\therefore \mathbb{P}\{\tau_* \in \widehat{\Theta}_A(e_*)\} \geq 1 - \sum_{\alpha \in \mathcal{W}_A(e_\star)} \zeta_\alpha \geq 1 - \zeta. \tag{A.62}$$

Summing the allocated error probabilities yields the simultaneous coverage claim in (A.62). For non-identification, let K be the closed design-compatible archive hull. If $\mathrm{dist}(m_\star, K) \geq r_0$, the function $g(m) = \mathrm{dist}(m, K)$ is 1-Lipschitz. The triangle inequality gives

$$\forall m, m', \forall k \in K \ \ \|m - k\| \leq \|m - m'\| + \|m' - k\| \tag{A.63}$$

The triangle inequality in (A.63), followed by taking infima over k, gives a one-sided distance comparison. Repeating the argument with m and m' exchanged gives the reverse comparison, hence

$$|d(m) - d(m')| \leq \|m - m'\|. \tag{A.64}$$

Thus (A.64) proves that $g(m) = \mathrm{dist}(m, K)$ is 1-Lipschitz. We now construct two effect curves using the truncated distance.

$$\mu_0(m) = 0, \mu_1(m) = a \min\{d(m), r_0\}, a = \min\{L, \frac{M}{r_0}\} \tag{A.65}$$

The definition of the two surfaces is given in (A.65).

$$\because |\mu_0(m) - \mu_0(m')| = 0 \le L\|m - m'\|, |\mu_0(m)| = 0 \le M \ \therefore \mu_0 \in \mathcal{L}(L, M) \tag{A.66}$$

The zero surface satisfies the Lipschitz and boundedness requirements as verified in (A.66). For the second surface, consider the truncated distance map:

$$\begin{aligned} |\mu_1(m) - \mu_1(m')| &= a \left|\min\{d(m), r_0\} - \min\{d(m'), r_0\}\right| \\ &\le a\|m - m'\| \\ &\le L\|m - m'\|, \end{aligned} \tag{A.67}$$

The Lipschitz calculation is (A.67). The range and boundedness checks are given next.

$$\because 0 \le \min\{d(m), r_0\} \le r_0, \therefore 0 \le \mu_1(m) \le ar_0. \tag{A.68}$$

$$\because ar_0 = \min\{Lr_0, M\} \le M \ \therefore |\mu_1(m)| \le M \therefore \mu_1 \in \mathcal{L}(L, M). \tag{A.69}$$

The range bound in (A.68) and the boundedness calculation in (A.69) establish that the second surface belongs to the admissible class. Since every archived mechanism lies in K, both surfaces agree on the archive, as shown in

$$\forall m_j \in K, d(m_j) = 0, \therefore \mu_0(m_j) = \mu_1(m_j) = 0 \tag{A.70}$$

Thus (A.70) verifies agreement of the two surfaces on all archived mechanisms. However, at the target the distance satisfies

$$d(m_\star) = \mathrm{dist}(m_\star, K) \ge r_0, \qquad \min\{d(m_\star), r_0\} = r_0.$$

$$\begin{aligned} |\mu_1(m_*) - \mu_0(m_*)| = ar_0 = r_0 \min\{L, \frac{M}{r_0}\} &= \min\{Lr_0, M\} \\ &\ge c_0 \min\{Lr_0, M\} \end{aligned} \tag{A.71}$$

At the unsupported target, the distance reaches the truncation level, so the effect separation is the value computed in (A.71). This proves the claimed partial-identification diameter lower bound.

A.9 Proof of Theorem 5.5

Proof strategy. The minimax lower bound is obtained from two independent submodels: a geometric pair of indistinguishable surfaces and a Gaussian two-point testing problem. The two submodels correspond to the two terms in the minimax statement (5.6). The geometric construction is developed in (A.72)–(A.83); the Gaussian testing construction is developed in (A.84)–(A.92). Let $R = \underset{\widehat{T}}{inf} \underset{\mu \in \mathcal{F}(L,M)}{sup} E_\mu \left|\widehat{T} - \mu(m_\star)\right|$ We prove two lower bounds and take their maximum. For the geometric bound, let $K = \mathrm{conv}\{m_j : j \in C\}$ and $d_\star = \mathrm{dist}(m_\star, K)$. If $d_\star = 0$, the geometric statement is trivial,

$$\because Ld_\star \wedge M = 0, \therefore R \ge 0 = Ld_\star \wedge M \tag{A.72}$$

If $d_\star > 0$, define $g(m) = \mathrm{dist}(m, K)$, which is 1-Lipschitz and equals zero on K. Let

$$a = \frac{1}{4}\{Ld_\star \wedge M\}, \qquad \mu_+(m) = a\,\frac{g(m) \wedge d_\star}{d_\star}, \qquad \mu_-(m) = -a\,\frac{g(m) \wedge d_\star}{d_\star}. \tag{A.73}$$

The two signed surfaces and their amplitude are defined in (A.73).

$$\because 0 \leq g(m) \wedge d_\star \leq d_\star \therefore |\mu_+(m)| \leq a \leq \frac{M}{4} \leq M, |\mu_-(m)| \leq a \leq \frac{M}{4} \leq M \tag{A.74}$$

The range check in (A.74) verifies the imposed bound M.

$$\begin{aligned} \left|\mu_+(m) - \mu_+(m')\right| &= \frac{a}{d_\star}\left|g(m) \wedge d_\star - g(m') \wedge d_\star\right| \\ &\leq \frac{a||m-m'||}{d_\star} \\ &\leq \frac{L}{4}||m-m'|| \\ &\leq L||m-m'|| \end{aligned} \tag{A.75}$$

The Lipschitz calculation in (A.75) verifies membership in the admissible function class. Thus μ_+ is L-Lipschitz. The same calculation, with the sign reversed, shows that μ_- is also L-Lipschitz.

Both functions belong to $\mathcal{F}(L, M)$ up to the universal constant absorbed by a, and both equal zero at every archived mechanism m_j. Hence the induced distributions of $(Z_j)_{j \in C}$ are identical under μ_+ and μ_-. At the target, however,

$$\because g(m_\star) = \mathrm{dist}(m_\star, K) = d_\star, \therefore g(m_\star) \wedge d_\star = d_\star \tag{A.76}$$

At the target, the truncated distance takes the value stated in (A.76).

$$\therefore \mu_+(m_\star) = a, \mu_-(m_\star) = -a, \because \mu_+, \mu_- \in \mathcal{F}(L, M) \tag{A.77}$$

Consequently, the target values are the opposite signs in (A.77).

$$\therefore \sup_{\mu \in \mathcal{F}(L,M)} E_\mu \left|\widehat{T} - \mu(m_\star)\right| \geq max\{E_{\mu_+}\left|\widehat{T} - a\right|, E_{\mu_-}\left|\widehat{T} + a\right|\} \tag{A.78}$$

The minimax supremum is bounded below by the two-point maximum in (A.78). μ_+, μ_- have the same distribution Q,

$$\because \left|\widehat{T} + a\right| + \left|\widehat{T} - a\right| \geq 2a \tag{A.79}$$

The pointwise absolute-value inequality used here is (A.79).

$$\therefore \frac{E_Q\left|\widehat{T} + a\right| + E_Q\left|\widehat{T} - a\right|}{2} \geq a \tag{A.80}$$

Averaging the two risks gives (A.80).

$$\therefore max\{E_Q\left|\widehat{T} + a\right|, E_Q\left|\widehat{T} - a\right|\} \geq a \tag{A.81}$$

At least one of the two risks must therefore satisfy the bound in (A.81).

$$\therefore \sup_{\mu \in \mathcal{F}(L,M)} E_\mu \left|\widehat{T} - \mu(m_\star)\right| \geq max\{E_{\mu_+}\left|\widehat{T} - a\right|, E_{\mu_-}\left|\widehat{T} + a\right|\} \geq a \tag{A.82}$$

Combining the two-point comparison with (A.82) yields the geometric minimax lower bound.

$$\therefore R \geq a = \frac{1}{4}\{Ld_\star \wedge M\} \tag{A.83}$$

Thus the geometric part gives the lower bound in (A.83), namely at least $c(Ld_\star \wedge M)$.

For the statistical bound, restrict to the constant submodel $\mu_t(m) = t$, $t \in [-M, M]$. Under t, the observations are independent Gaussians $Z_j \sim \mathsf{N}(t, s_j^2)$. Let $I = \sum_{j \in C} s_j^{-2}$. Choose $t_+ = \delta$ and $t_- = -\delta$, where $\delta = c_1(I^{-1/2} \wedge M)$ and $c_1 > 0$ is small. The two-point risk reduction and the two-point reduction and the two Gaussian product laws are displayed in (A.84) and (A.85); the choice of δ gives the information bound in (A.86).

$$R \geq \inf_{\widehat{T}} max\{E_{t_+}\left|\widehat{T} - \delta\right|, E_{t_-}\left|\widehat{T} + \delta\right|\} \tag{A.84}$$

$$P_{t_+} = \bigotimes_{j \in C} \mathsf{N}(\delta, s_j^2), \qquad P_{t_-} = \bigotimes_{j \in C} \mathsf{N}(-\delta, s_j^2). \tag{A.85}$$

$$\because \delta \leq c_1 I^{-1/2} \ \therefore 2\delta^2 I \leq 2c_1^2. \tag{A.86}$$

The Kullback–Leibler divergence between the two product measures is computed in

$$\mathrm{KL}(P_{t_+}, P_{t_-}) = \frac{1}{2} \sum_{j \in C} \frac{(2\delta)^2}{s_j^2} = 2\delta^2 I \leq 2c_1^2, \tag{A.87}$$

The resulting KL bound in (A.87) is converted to total variation by Pinsker's inequality in (A.88). Let $\mathrm{TV}(P, Q) := \sup_A |P(A) - Q(A)|$, using Pinsker inequality,

$$\mathrm{TV}(P_{t_+}, P_{t_-}) \leq \sqrt{\frac{1}{2}\,\mathrm{KL}(P_{t_+}, P_{t_-})} \leq c_1. \tag{A.88}$$

$\forall \widehat{T}$ define $A = \{\widehat{T} \geq 0\}$,
If the real parameter is $t_+ = \delta, \widehat{T} < 0$,thus $|\widehat{T} - \delta| \geq \delta, \therefore \mathbb{E}_{t_+}|\widehat{T} - \delta| \geq \delta P_{t_+}(A^c)$. Similarly, if the real parameter is $t_- = -\delta, \widehat{T} \geq 0$,thus $|\widehat{T} + \delta| \geq \delta, \therefore \mathbb{E}_{t_-}|\widehat{T} + \delta| \geq \delta P_{t_-}(A)$.

$$\begin{aligned} \therefore \mathbb{E}_{t_+}|\widehat{T} - \delta| + \mathbb{E}_{t_-}|\widehat{T} + \delta| &\geq \delta\{P_{t_+}(A^c) + P_{t_-}(A)\} \\ &= \delta(1 - P_{t_+}(A) + P_{t_-}(A)) \\ &\geq \delta(1 - TV(P_{t_+}, P_{t_-})) \geq \delta(1 - c_1) \end{aligned} \tag{A.89}$$

The testing event $A = \{\widehat{T} \geq 0\}$ then gives the two error contributions combined in (A.89); averaging them yields the maximum-risk bound in (A.90).

$$\therefore \max\left\{\mathbb{E}_{t_+}|\widehat{T} - \delta|,\, \mathbb{E}_{t_-}|\widehat{T} + \delta|\right\} \geq (\frac{1 - c_1}{2})\delta. \tag{A.90}$$

$$\therefore R \geq \frac{c_1(1 - c_1)}{2}(I^{-\frac{1}{2}} \wedge M) \tag{A.91}$$

The statistical lower bound is therefore the order statement in (A.91). Le Cam's two-point lemma justifies the same conclusion in standard minimax terminology. Combining the geometric and statistical submodels yields the target bound (5.6). Let $c = \min\{\frac{1}{4}, \frac{c_1(1-c_1)}{2}\}$, we have

$$R \geq c\left[\{Ld_\star\} \wedge M \ \vee \{\sum_{j \in C} s_j^{-2}\}^{-\frac{1}{2}} \wedge M\right] \tag{A.92}$$

The final constant choice in (A.92) takes the maximum of the geometric and statistical lower bounds, completing the proof of Theorem 5.5.

A.10 Proof of Theorem 5.6

Proof strategy. We prove the greedy guarantee through a one-step marginal-gain inequality, a deterministic recurrence for the remaining optimality gap, and an explicit solution of that recurrence.

The objective is the bridge value in (5.7), with the diameter formulation supplied at the end of the proof. The proof introduces the true marginal in (A.93), conditions on the uniform estimation event in (A.94), derives the greedy one-step inequality in (A.105), and solves the resulting gap recurrence in (A.110).

We prove the result for $B \geq 1$ and $0 < \gamma \leq 1$, which is the standard range of the weak-submodularity parameter. The case $B = 0$ is trivial because $S_0 = \varnothing$. For a set $S \subseteq \mathcal{B}$ and a candidate $u \in \mathcal{B} \setminus S$, define the true marginal value of u by

$$\Delta(u \mid S) := F(S \cup \{u\}) - F(S). \tag{A.93}$$

The definition in (A.93) is the marginal gain whose noisy version is optimized by the algorithm. Let $\widehat{\Delta}(u \mid S)$ denote the marginal value used by the algorithm. The assumed uniform estimation condition is

$$\left|\widehat{\Delta}(u \mid S) - \Delta(u \mid S)\right| \leq \varepsilon_{\text{est}} \qquad \text{for every eligible pair } (u, S). \tag{A.94}$$

The event in (A.94) is the uniform error certificate used throughout the greedy comparison. If the estimated marginal values are random, all arguments below are conditional on that event; consequently, the final conclusion holds with the probability of the event.

Set $S_0 = \varnothing$. At step $b \in \{1, \dots, B\}$, the greedy algorithm chooses

$$u_b \in \arg\max_{u \in \mathcal{B} \setminus S_{b-1}} \widehat{\Delta}(u \mid S_{b-1}), \qquad S_b = S_{b-1} \cup \{u_b\}. \tag{A.95}$$

The selected set is updated according to the greedy rule in (A.95). Fix one step b, and define the part of the optimal set that has not yet been selected:

$$U_b := S^\star \setminus S_{b-1}. \tag{A.96}$$

The set U_b in (A.96) contains the optimal candidates not yet selected by the greedy sequence. If $U_b = \varnothing$, then $S^\star \subseteq S_{b-1}$. Monotonicity of F gives

$$F(S_{b-1}) \geq F(S^\star), \tag{A.97}$$

The monotonicity conclusion in (A.97) handles the case in which no optimal candidate remains. so the desired lower bound has already been reached at this step. We therefore consider the nontrivial case $U_b \neq \varnothing$.

Apply weak submodularity with

$$S = S_{b-1}, \qquad T = S_{b-1} \cup S^\star. \tag{A.98}$$

The substitution $S = S_{b-1}$ and $T = S_{b-1} \cup S^\star$ is recorded in (A.98). Then

$$\begin{aligned} \sum_{u \in U_b} \Delta(u \mid S_{b-1}) &= \sum_{u \in T \setminus S} \{F(S \cup \{u\}) - F(S)\} \\ &\geq \gamma\{F(S_{b-1} \cup S^\star) - F(S_{b-1})\}. \end{aligned} \tag{A.99}$$

Weak submodularity applied at (A.98) gives the marginal sum in (A.99); monotonicity then supplies (A.100). Because F is monotone and $S^\star \subseteq S_{b-1} \cup S^\star$,

$$F(S_{b-1} \cup S^\star) \geq F(S^\star). \tag{A.100}$$

Therefore, the summed marginal inequality is recorded as (A.101).

$$\sum_{u \in U_b} \Delta(u \mid S_{b-1}) \geq \gamma\{F(S^\star) - F(S_{b-1})\}. \tag{A.101}$$

The set $S^\star$ has at most B elements, so $|U_b| \leq B$. The average of the marginal values in (A.101) is therefore at least

$$\frac{\gamma}{B}\{F(S^\star) - F(S_{b-1})\}. \tag{A.102}$$

Dividing the sum bound (A.101) by the cardinality bound produces the average marginal in (A.102). Consequently, there exists an element $u_b^\star \in U_b$ satisfying the good-candidate bound (A.103):

$$\Delta(u_b^\star \mid S_{b-1}) \geq \frac{\gamma}{B}\{F(S^\star) - F(S_{b-1})\}. \tag{A.103}$$

We next compare the true marginal gain of the selected element u_b with that of $u_b^\star$. The first inequality below uses (A.94); the second uses the definition of the greedy choice; the third uses the same uniform error bound again:

$$\begin{aligned} \Delta(u_b \mid S_{b-1}) &\geq \widehat{\Delta}(u_b \mid S_{b-1}) - \varepsilon_{\text{est}} \\ &\geq \widehat{\Delta}(u_b^\star \mid S_{b-1}) - \varepsilon_{\text{est}} \\ &\geq \Delta(u_b^\star \mid S_{b-1}) - 2\varepsilon_{\text{est}}. \end{aligned} \tag{A.104}$$

The three comparisons in (A.104) are, respectively, the uniform-error lower bound, greedy maximization, and the second uniform-error bound. Combining this display with (A.103) yields

$$\begin{aligned} F(S_b) - F(S_{b-1}) &= \Delta(u_b \mid S_{b-1}) \\ &\geq \frac{\gamma}{B}\{F(S^\star) - F(S_{b-1})\} - 2\varepsilon_{\text{est}}. \end{aligned} \tag{A.105}$$

This is the key one-step progress inequality: the greedy step captures at least a γ/B fraction of the remaining optimal value, up to the error $2\varepsilon_{\text{est}}$. This is also the theorem-level one-step inequality (A.105).

Define the remaining optimality gap after b steps by

$$G_b := F(S^\star) - F(S_b), \qquad b = 0, 1, \ldots, B. \tag{A.106}$$

The gap variable in (A.106) measures the value still missing from the optimal bridge set. Since $S_0 = \varnothing$ and $F(\varnothing) = 0$,

$$G_0 = F(S^\star). \tag{A.107}$$

Thus the initial gap is the optimal value itself, as stated in (A.107). Rearranging (A.105) gives

$$G_b \leq \left(1 - \frac{\gamma}{B}\right) G_{b-1} + 2\varepsilon_{\text{est}}. \tag{A.108}$$

Rearranging the one-step inequality produces the gap recursion in (A.108).

For completeness, we solve this recurrence explicitly. Let

$$q := 1 - \frac{\gamma}{B}. \tag{A.109}$$

The shorthand q in (A.109) is the contraction factor for the deterministic gap recursion. Because $0 < \gamma \leq 1$ and $B \geq 1$, $0 \leq q < 1$. We claim by induction that, for every $b \in \{0, \ldots, B\}$,

$$G_b \leq q^b G_0 + 2\varepsilon_{\text{est}} \sum_{k=0}^{b-1} q^k, \tag{A.110}$$

The induction claim is the explicit recurrence solution in (A.110). where the sum is interpreted as zero when $b = 0$. For $b = 0$, (A.110) is an equality. Suppose it holds for $b - 1$. Applying (A.108) and then the induction hypothesis gives

$$\begin{aligned} G_b &\leq q G_{b-1} + 2\varepsilon_{\text{est}} \\ &\leq q \left(q^{b-1} G_0 + 2\varepsilon_{\text{est}} \sum_{k=0}^{b-2} q^k \right) + 2\varepsilon_{\text{est}} \\ &= q^b G_0 + 2\varepsilon_{\text{est}} \sum_{k=0}^{b-1} q^k. \end{aligned} \tag{A.111}$$

The induction step in (A.111) verifies that the claimed solution is preserved from $b - 1$ to b. Thus (A.110) holds for every b, and in particular

$$G_B \leq q^B F(S^\star) + 2\varepsilon_{\text{est}} \sum_{k=0}^{B-1} q^k. \tag{A.112}$$

At the final step, the recurrence gives the bound in (A.112).

We now bound the two terms on the right-hand side. The elementary inequality $1 - x \leq e^{-x}$ for $x \geq 0$ gives

$$q^B = \left(1 - \frac{\gamma}{B}\right)^B \leq e^{-\gamma}. \tag{A.113}$$

The exponential contraction estimate is (A.113). Also, since $1 - q = \gamma/B$,

$$\sum_{k=0}^{B-1} q^k = \frac{1 - q^B}{1 - q} = \frac{B(1 - q^B)}{\gamma} \leq \frac{B}{\gamma}. \tag{A.114}$$

The geometric-series estimate is (A.114); together with (A.113) it implies Consequently,

$$G_B \leq e^{-\gamma} F(S^\star) + \frac{2B\varepsilon_{\text{est}}}{\gamma}. \tag{A.115}$$

Hence the remaining gap is bounded as in (A.115). Since $G_B = F(S^\star) - F(S_B)$, subtracting this inequality from $F(S^\star)$ gives

$$\begin{aligned} F(S_B) &= F(S^\star) - G_B \\ &\geq (1 - e^{-\gamma}) F(S^\star) - \frac{2B\varepsilon_{\text{est}}}{\gamma}. \end{aligned} \tag{A.116}$$

Equation (A.116) is the claimed greedy bridge-design guarantee in value-of-information units.

Finally, substitute the definition The value function used to translate this guarantee into an expected diameter is defined in (A.117).

$$F(S) = \operatorname{diam}\{\Theta_A(e_\star)\} - \mathbb{E}[\operatorname{diam}\{\Theta_{A\cup S}(e_\star)\}]\,. \tag{A.117}$$

Rearranging the preceding inequality gives Substituting (A.116) and (A.117) gives the expected-diameter inequality in

$$\begin{aligned}\mathbb{E}[\operatorname{diam}\{\Theta_{A\cup S_B}(e_\star)\}] \le\ & \operatorname{diam}\{\Theta_A(e_\star)\} - (1-e^{-\gamma})F(S^\star) \\ & + \frac{2B\varepsilon_{\text{est}}}{\gamma},\end{aligned} \tag{A.118}$$

which is the equivalent expected-diameter formulation in (A.118) and the second form of (5.7).

B Appendix: Experimental Details

This appendix records eight experimental blocks supporting Section 6. The paper-facing numbers and tables are generated from the repository `https://github.com/LoverDK/experiment` at release commit `96fba35`. The run scripts write fixed CSV/JSON results, and the paper builder converts those committed results into tables and vector figures without resampling Monte Carlo targets. The first two subsections discharge the two implementation details explicitly deferred from Section 6: the complete synthetic data-generating process and the formal estimator ablations.

B.1 Synthetic DGP and Assumption Validity

Mechanisms and support. Each archive experiment has mechanism

$$m = (s_1, s_2, h, q) \in [-1, 1]^4,$$

where s_1, s_2 are semantic coordinates, h is a hidden moderator, and q is a design/context coordinate. Eight archive mechanisms are sampled independently and uniformly on the mechanism cube. A supported target is generated from independent simplex weights,

$$m_{\text{sup}} = \sum_{j=1}^{8} \omega_j m_j, \qquad (\omega_1, \dots, \omega_8) \sim \text{Dirichlet}(1, \dots, 1).$$

Support deterioration moves the target toward the fixed in-domain anchor $a = (1, -1, 1, -1)$:

$$m_\star(\rho) = (1-\rho)m_{\text{sup}} + \rho a, \qquad \rho \in \{0, 0.10, 0.25, 0.60, 0.80\}.$$

The target remains inside the compact smoothness domain while its distance from the archive hull changes. The configuration constants governing this construction and the optimizer are collected in Table 5.

Effect surface and potential outcomes. The exact effect surface is

$$\mu(m) = 1.15\sin(1.1s_1) + 0.65s_2 + 1.10h + 0.45s_1h - 0.28q^2 + 0.25\cos(s_2 + q). \tag{B.1}$$

The analytic constants supplied to the certificate are $L = 2.61$, $H = 1.80$, $L_h = 1.55$, and $|\mu(m)| \le 3.88$. For unit ℓ,

$$X_\ell \sim \mathsf{N}(0, I_2), \qquad g(X_\ell) = 0.50X_{\ell 1} - 0.30X_{\ell 2}, \tag{B.2}$$
$$Y_\ell(0) = g(X_\ell) + \epsilon_{\ell 0}, \qquad Y_\ell(1) = g(X_\ell) + \mu(m) + \epsilon_{\ell 1}, \tag{B.3}$$
$$A_\ell \sim \text{Bernoulli}(0.5), \qquad Y_\ell = A_\ell Y_\ell(1) + (1 - A_\ell)Y_\ell(0), \tag{B.4}$$

where the errors are independent $\mathsf{N}(0, 1)$. The public representation is $r = (s_1, s_2, \widetilde{h}, q)$, with

$$\widetilde{h} = \text{clip}\{h + U, -1, 1\}, \qquad U \sim \text{Unif}[-0.10, 0.10].$$

The nominal moderator-sensitivity radius is 0.20, and the radius-sensitivity experiments use 0.40 and larger values in the one-factor sweeps.

Experiment-level estimator and uncertainty. The nuisance regressions are known in this controlled DGP. The AIPW score stored by archive object i is

$$\phi_{i\ell} = \mu_1(X_{i\ell}) - \mu_0(X_{i\ell}) + \frac{A_{i\ell}\{Y_{i\ell} - \mu_1(X_{i\ell})\}}{0.5} - \frac{(1 - A_{i\ell})\{Y_{i\ell} - \mu_0(X_{i\ell})\}}{0.5}. \tag{B.5}$$

The stored effect and standard-error objects are

$$\widehat{\tau}_i = \frac{1}{n_i}\sum_{\ell=1}^{n_i} \phi_{i\ell}, \qquad \widehat{v}_i = \frac{1}{n_i - 1}\sum_{\ell=1}^{n_i}(\phi_{i\ell} - \widehat{\tau}_i)^2, \qquad s_i^2 = \widehat{v}_i/n_i. \tag{B.6}$$

The nuisance remainder is therefore zero by construction, while finite-sample noise, representation residual, curvature, and hidden-moderator uncertainty remain present.

Assumption checks. Before any performance comparison, the implementation validates the five conditions used by the theory. Random treatment assignment, consistency, positivity, and a common estimand establish the experiment-level identification and design-normalization requirements in Assumptions 3.1–3.2. Analytic gradient and Hessian bounds verify Assumption 3.3. Independent archive random streams, the AIPW score variance, and zero oracle-nuisance remainder instantiate the uncertainty certificate in Assumption 3.4. Finally, the proxy construction satisfies $|h - \widetilde{h}| \le 0.10$, while the declared radius 0.20 and $L_h = 1.55$ bound the corresponding effect variation required by Assumption 3.5. Automated tests re-evaluate the analytic derivatives, randomization, support residual, and true-versus-proxy discrepancy. These checks validate the declared simulation model; they are not additional evidence about estimator performance.

Operational representation versus latent support. The latent mechanism m is used in the synthetic generator to define the effect surface and, after prediction, to evaluate known-truth support and reconstruction. The deployable atlas does not observe m: it retrieves and weights experiments using the observed representation $r = (s_1, s_2, \widetilde{h}, q)$ and covers the unobserved part of $m - r$ through the declared hidden-moderator certificate $R_{\text{hid}}(\alpha)$. The evaluation-only latent oracle is the sole row allowed to replace r by m; it is never supplied to retrieval, weighting, release, or bridge selection. Thus the oracle measures the value of latent support, whereas the reported Causal ATLAS results test the operational representation and its stated moderator certificate.

Table 5: Nominal synthetic configuration and optimization constants.

Quantity	Value	Quantity	Value
Archive experiments	8	Units per experiment	400
Target units	400	Treatment probability	0.50
Overlap lower bound	0.10	Outcome noise SD	1.00
Proxy half-width	0.10	Moderator radius	0.20
Scientific tolerance	1.65	Confidence error ζ	0.05
Variance penalty λ_σ	1.00	Hidden penalty λ_h	1.00
Learning rate	0.25	Weight iterations	800
Convergence tolerance	10^{-10}	Candidate cap	No cap

B.2 Formal Stress Scenarios and Estimator Ablations

Formal benchmark and ablations. The formal runner uses base seeds 20260811–20260813, 100 generated targets per seed, and the six scenarios in the committed metadata: nominal, semantic shifts 0.10 and 0.25, moderator radius 0.40, and sample sizes 100 and 1000. The complete nominal comparison, including the no-variance-penalty and top-4-candidate ablations, is in Table 6; the two stress-table inputs contain all 42 scenario–method rows, reported in Tables 7 and 8.

Table 6: Complete nominal formal benchmark and estimator ablations under a separately fixed formal-stage protocol. This formal-stage protocol differs from the shared-target diagnostic protocol used for Table 1: it pools its own seeds and generated targets and therefore targets a different numerical diagnostic. Row-wise values need not match Table 1 and should not be read as inconsistencies. Reported errors are conditional on release for rejectable rows.

Method	Release	MAE	RMSE	Sign	Coverage	Width
Causal ATLAS	0.463	0.111	0.138	0.928	1.000	3.339
ATLAS, no rejection	1.000	0.139	0.170	0.893	1.000	3.339
No variance penalty	0.440	0.112	0.140	0.924	1.000	3.339
Top-4 candidates	0.320	0.123	0.155	0.906	1.000	3.626
Semantic forced	1.000	0.224	0.282	0.827	1.000	4.477
Nearest semantic	1.000	0.456	0.568	0.787	1.000	4.809
Global mean	1.000	0.282	0.362	0.780	1.000	5.224

Table 7: Formal sensitivity benchmark across the stated DGP change. An empty MAE indicates complete rejection rather than a failed evaluation.

Scenario	Method	Release	MAE	RMSE	Sign	Coverage	Width
Semantic shift 0.10	Causal ATLAS	0.390	0.126	0.155	0.940	1.000	3.410
Semantic shift 0.10	ATLAS, no rejection	1.000	0.158	0.191	0.903	1.000	3.410
Semantic shift 0.10	No variance penalty	0.383	0.123	0.152	0.939	1.000	3.407
Semantic shift 0.10	Top-4 candidates	0.263	0.130	0.162	0.937	1.000	3.728
Semantic shift 0.10	Semantic forced	1.000	0.298	0.360	0.770	1.000	4.680
Semantic shift 0.10	Nearest semantic	1.000	0.488	0.596	0.770	1.000	4.845
Semantic shift 0.10	Global mean	1.000	0.365	0.445	0.703	1.000	5.423
Semantic shift 0.25	Causal ATLAS	0.283	0.158	0.191	0.953	1.000	3.839
Semantic shift 0.25	ATLAS, no rejection	1.000	0.231	0.281	0.883	1.000	3.839
Semantic shift 0.25	No variance penalty	0.280	0.156	0.190	0.952	1.000	3.834
Semantic shift 0.25	Top-4 candidates	0.210	0.153	0.189	0.952	1.000	3.999
Semantic shift 0.25	Semantic forced	1.000	0.458	0.514	0.717	1.000	5.583
Semantic shift 0.25	Nearest semantic	1.000	0.541	0.666	0.760	1.000	4.963
Semantic shift 0.25	Global mean	1.000	0.569	0.625	0.597	1.000	6.498
Moderator radius 0.40	Causal ATLAS	0.000	–	–	–	1.000	4.579
Moderator radius 0.40	ATLAS, no rejection	1.000	0.139	0.170	0.893	1.000	4.579
Moderator radius 0.40	No variance penalty	0.000	–	–	–	1.000	4.579
Moderator radius 0.40	Top-4 candidates	0.000	–	–	–	1.000	4.866
Moderator radius 0.40	Semantic forced	1.000	0.224	0.282	0.827	1.000	5.717
Moderator radius 0.40	Nearest semantic	1.000	0.456	0.568	0.787	1.000	6.049
Moderator radius 0.40	Global mean	1.000	0.282	0.362	0.780	1.000	6.464

Table 8: Formal sensitivity benchmark across the stated DGP change. An empty MAE indicates complete rejection rather than a failed evaluation.

Scenario	Method	Release	MAE	RMSE	Sign	Coverage	Width
100 units	Causal ATLAS	0.230	0.123	0.151	0.870	1.000	3.537
100 units	ATLAS, no rejection	1.000	0.150	0.185	0.880	1.000	3.537
100 units	No variance penalty	0.247	0.125	0.151	0.892	1.000	3.509
100 units	Top-4 candidates	0.173	0.161	0.192	0.904	1.000	3.868
100 units	Semantic forced	1.000	0.232	0.290	0.830	1.000	4.614
100 units	Nearest semantic	1.000	0.523	0.641	0.760	1.000	5.155
100 units	Global mean	1.000	0.282	0.367	0.763	1.000	5.363
1000 units	Causal ATLAS	0.500	0.113	0.136	0.940	1.000	3.271
1000 units	ATLAS, no rejection	1.000	0.139	0.167	0.903	1.000	3.271
1000 units	No variance penalty	0.493	0.112	0.137	0.932	1.000	3.277
1000 units	Top-4 candidates	0.397	0.134	0.166	0.891	1.000	3.537
1000 units	Semantic forced	1.000	0.224	0.284	0.820	1.000	4.403
1000 units	Nearest semantic	1.000	0.463	0.568	0.753	1.000	4.625
1000 units	Global mean	1.000	0.280	0.363	0.773	1.000	5.157

The two ablations isolate different parts of the composition rule. Removing the variance penalty sets $\lambda_\sigma = 0$ while leaving retrieval, the hidden-moderator penalty, certification, and release unchanged. Restricting retrieval to the four nearest compatible candidates changes only the candidate set presented to the weight optimizer. In the nominal design, full ATLAS releases 0.463 of targets with released-target MAE 0.111. Removing the variance penalty changes these values only to 0.440 and 0.112, whereas the top-four restriction lowers release to 0.320, raises MAE to 0.123, and increases mean interval width from 3.339 to 3.626. Thus candidate richness matters more than variance regularization in this particular homoskedastic design; the experiment does not imply that variance regularization is generally unimportant. A no-hidden-penalty ablation is omitted because the nominal archive radii are equal, making that penalty constant over the simplex and therefore uninformative as a weight ablation.

The empty MAE cells in the radius-0.40 rows are complete rejection. Increasing the hidden-moderator radius from 0.20 to 0.40 produces a release rate of zero for certified ATLAS while forced baselines continue to publish. Increasing the sample size from 100 to 1,000 raises release from 0.230 to 0.500, while moving the semantic shift from zero to 0.25 lowers release from 0.463 to 0.283 and raises released-target MAE from 0.111 to 0.158. These responses separate sampling precision, representation mismatch, and declared moderator uncertainty. The formal runner and the target-level certificate diagnostic are separately fixed protocols; the latter is the source of the common-target Table 1 because it retains every target-level certificate component and the evaluation-only oracle.

B.3 Representation Sensitivity and One-Factor Robustness

The repository adds a 5×5 grid with five hidden-shift levels from 0 to 0.8 and five proxy-uncertainty levels from 0.05 to 0.40. Each cell uses the same three base seeds and 100 repetitions per seed. The representation advantage compares semantic forced MAE with ATLAS no-rejection MAE; the selection gain is reported separately and is not treated as an additive causal decomposition. Table 9 gives the full two-factor robustness diagnostic; the one-factor sweeps are retained as a separate robustness check in the experiment archive.

Table 9: Full 5×5 representation-sensitivity grid. Δ_{rep} is semantic-forced MAE minus ATLAS no-rejection MAE; selection gain is a separate conditional-risk diagnostic.

Hidden shift	Proxy uncertainty	Release	ATLAS no-rej. MAE	Semantic MAE	Δ_{rep}	Selection gain
0.00	0.05	0.967	0.131	0.256	0.125	0.003
0.00	0.10	0.887	0.133	0.256	0.123	0.009
0.00	0.20	0.457	0.140	0.256	0.116	0.027
0.00	0.30	0.043	0.151	0.256	0.105	0.067
0.00	0.40	0.003	0.165	0.256	0.091	0.063
0.20	0.05	0.943	0.142	0.372	0.230	0.006
0.20	0.10	0.850	0.145	0.372	0.227	0.013
0.20	0.20	0.417	0.154	0.372	0.218	0.032
0.20	0.30	0.057	0.167	0.372	0.205	0.050
0.20	0.40	0.000	0.183	0.372	0.188	–
0.40	0.05	0.853	0.186	0.558	0.372	0.021
0.40	0.10	0.703	0.189	0.558	0.369	0.034
0.40	0.20	0.327	0.198	0.558	0.361	0.074
0.40	0.30	0.083	0.211	0.558	0.348	0.085
0.40	0.40	0.000	0.227	0.558	0.331	–
0.60	0.05	0.600	0.267	0.772	0.505	0.087
0.60	0.10	0.433	0.270	0.772	0.502	0.110
0.60	0.20	0.230	0.278	0.772	0.494	0.131
0.60	0.30	0.050	0.288	0.772	0.484	0.160
0.60	0.40	0.000	0.302	0.772	0.471	–
0.80	0.05	0.373	0.384	0.990	0.606	0.162
0.80	0.10	0.260	0.386	0.990	0.603	0.179
0.80	0.20	0.093	0.392	0.990	0.597	0.211
0.80	0.30	0.013	0.402	0.990	0.588	0.274
0.80	0.40	0.000	0.414	0.990	0.576	–

At proxy uncertainty 0.10, increasing the hidden shift from zero to 0.80 increases the full-representation advantage from 0.123 to 0.603, while the certified release rate falls from 0.887 to 0.260. This is the intended distinction between reconstruction and certification: a richer representation can improve a forced point prediction even as the certificate detects that fewer targets are safe to release. The one-factor sweeps vary semantic mismatch, hidden-moderator radius, experiment sample size, and scientific tolerance. They are robustness checks rather than additional claims about an optimal operating point. Higher sample size narrows statistical uncertainty, whereas a larger hidden radius or semantic mismatch increases the certificate and can reduce release even when the no-rejection point estimate is unchanged.

B.4 Certificate Decomposition, Calibration, and Misspecification

Certificate decomposition. Table 10 decomposes the certificate radius across released and rejected targets. The target-level diagnostic records the representation, curvature, hidden-moderator, bias, and statistical components for all 300 common targets. Among the 139 released targets, the mean curvature component is 0.759, compared with 1.131 among the 161 rejected targets. The hidden-moderator components are both 0.620, and the statistical components are 0.118 and 0.113, respectively. Rejection in this nominal DGP is therefore driven mainly by geometric support rather than by sampling noise. This component ranking is specific to the fixed DGP and is not a general claim that moderator or sampling uncertainty is negligible.

Table 10: Target-level certificate-component decomposition on the 300 common targets, reported separately for released and rejected targets.

Status	n	Representation	Curvature	Hidden	Statistical	Radius	Abs. error
Released	139	0.009	0.759	0.620	0.118	1.506	0.111
Rejected	161	0.006	1.131	0.620	0.113	1.870	0.163

The target-level certificate radius–realized-error diagnostic is shown in Figure 6; it is a descriptive diagnostic of release and rejection, not a pointwise calibration guarantee.

Selective thresholds. The risk–coverage experiment uses the same 300 nominal targets at every threshold and changes only the scientific tolerance. Release rates are 0.010, 0.500, and 0.943 at tolerances 1.25, 1.65, and 2.00, while released-target MAE changes from 0.074 to 0.116 and 0.133. The no-rejection endpoint has MAE 0.136. Empty operating points below 1.25 are retained in the source CSV but omitted from the compact table because conditional risk is undefined when no target is released; the retained nonempty frontier is summarized in Table 11.

Appendix B.4. Target-level certificate diagnostic

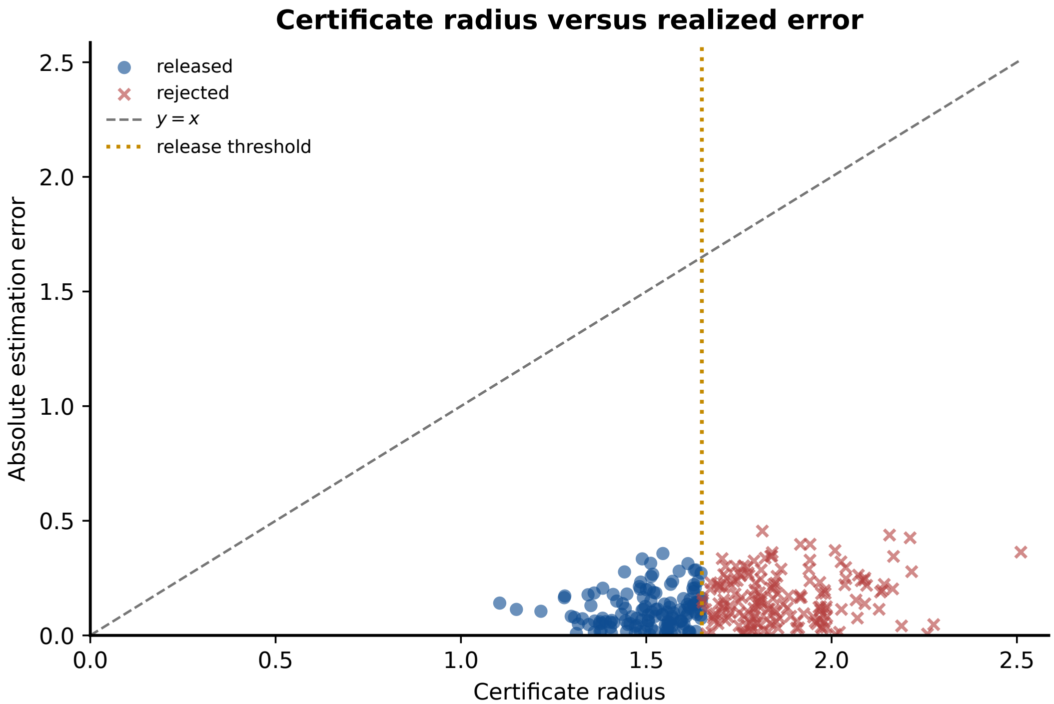

Figure 6: Target-level certificate radius versus realized absolute error for the common synthetic targets. Released and rejected targets are shown separately; the diagnostic is evaluation-only and does not replace the formal certificate guarantee.

Table 11: Complete risk–coverage threshold sweep on the same 300 nominal targets.

Threshold	Release	Count	Conditional MAE	RMSE	Coverage	Width
1.25	0.010	3	0.074	0.087	1.000	2.312
1.50	0.197	59	0.114	0.139	1.000	2.736
1.65	0.500	150	0.116	0.142	1.000	2.946
2.00	0.943	283	0.133	0.160	1.000	3.224
2.50	1.000	300	0.136	0.164	1.000	3.274
3.00	1.000	300	0.136	0.164	1.000	3.274
No rejection	1.000	300	0.136	0.164	1.000	3.274

Coverage–width grid. The confidence-level experiment evaluates nominal levels 0.80, 0.90, 0.95, and 0.975 under honest ATLAS, Wald-only, semantic forced, understated-smoothness, and no-hidden-moderator-inflation policies. Honest intervals have empirical coverage 1.000 throughout, with mean width increasing from 3.283 to 3.367. Wald-only coverage increases only from 0.200 to 0.337 and its width remains below 0.20, showing why coverage must be interpreted jointly with interval width. Table 12 reports these coverage–width diagnostics together with release rates.

Table 12: All confidence-level diagnostics. Coverage is always reported with width and the release rate.

Policy	Nominal	Release	Empirical coverage	Mean width
Honest ATLAS	0.800	0.467	1.000	3.283
Wald only	0.800	1.000	0.200	0.111
Semantic forced	0.800	1.000	1.000	4.520
Understated smoothness	0.800	1.000	1.000	1.406
No hidden inflation	0.800	0.997	1.000	2.043
Honest ATLAS	0.900	0.443	1.000	3.315
Wald only	0.900	1.000	0.240	0.143
Semantic forced	0.900	1.000	1.000	4.548
Understated smoothness	0.900	1.000	1.000	1.437
No hidden inflation	0.900	0.997	1.000	2.075
Honest ATLAS	0.950	0.403	1.000	3.342
Wald only	0.950	1.000	0.287	0.170
Semantic forced	0.950	1.000	1.000	4.571
Understated smoothness	0.950	1.000	1.000	1.464
No hidden inflation	0.950	0.997	1.000	2.102
Honest ATLAS	0.975	0.387	1.000	3.367
Wald only	0.975	1.000	0.337	0.194
Semantic forced	0.975	1.000	1.000	4.593
Understated smoothness	0.975	1.000	1.000	1.489
No hidden inflation	0.975	0.997	1.000	2.127

Failure boundary. The strong and severe mismatch scenarios deliberately set the smoothness bounds below their valid values. Coverage then falls to 0.833 and 0.723, respectively. This negative control demonstrates that a narrow numerical certificate has its intended interpretation only when the scientific upper bounds entering it are valid; the deliberate boundary is quantified in Table 13.

Table 13: Deliberate certificate misspecification boundary. Understated smoothness is a negative control that violates the declared certificate assumptions.

Scenario	Policy	Release	Released MAE	Coverage	Above tolerance
Strong mismatch	Certified ATLAS	0.043	0.235	1.000	0.000
Strong mismatch	No rejection	1.000	0.555	1.000	0.957
Strong mismatch	Understated bounds	1.000	0.555	0.833	0.000
Severe mismatch	Certified ATLAS	0.007	0.180	1.000	0.000
Severe mismatch	No rejection	1.000	0.741	1.000	0.993
Severe mismatch	Understated bounds	1.000	0.741	0.723	0.000

B.5 Partial Identification after Rejection

The partial-identification protocol uses seeds 20260911–20260913, 100 targets per seed, shift fractions 0, 0.25, 0.60, and 0.80, and at most four singleton archive certificates in the intersection. Duplicate weight certificates are removed, and the failure probability is divided across the distinct retained weights before their intervals are intersected. Empty intersections are recorded as mutually inconsistent certificates under Lemma 5.2; they are never assigned zero width. Synthetic truth is used only for evaluating coverage. Table 14 reports the complete partial-identification diagnostics, including the evaluation-only oracle-hull quantities.

Table 14: Complete partial-identification diagnostics. Oracle-hull distance and nonidentification separation are simulation-only evaluation quantities.

Scenario	Reject.	Nonempty	Coverage	PI diameter	Reference diameter	Reduction	Oracle hull	Separation
Nominal	0.517	1.000	1.000	3.646	3.752	0.027	0.000	0.000
Moderate	0.777	1.000	1.000	4.095	4.343	0.051	0.133	0.346
Strong	0.973	1.000	1.000	5.674	6.484	0.114	0.647	1.689
Severe	0.990	1.000	1.000	7.076	8.068	0.113	0.997	2.598

As support deteriorates, rejection increases from 0.517 to 0.990 and the mean diameter on rejected targets increases from 3.646 to 7.076. Every intersection is nonempty and every rejected-branch interval contains the simulated truth in the prespecified scenarios. Intersecting the valid weight-specific intervals reduces width relative to the single support-optimized reference by 2.74%, 5.14%, 11.43%, and 11.25% across the four scenarios. The widening set is therefore the intended representation of weaker identification, while the intersection uses multiple compatible archive restrictions to avoid unnecessary width.

B.6 Numerical Illustration of the Minimax Lower Bound

The minimax experiment is a numerical sanity check for Theorem 5.5 rather than a replacement for its proof. It evaluates the geometric and statistical two-point submodels under archive standard errors 0.35 and 1.20, hull distances 0, 0.25, 0.60, and 1.00, and 300 repetitions pooled over three seeds. The reported constructive lower bound is the maximum of the two component lower bounds; Table 15 presents the resulting numerical illustration without elevating it to a formal result.

Table 15: Numerical illustration of the two proof submodels in Theorem 5.5. The constructive lower bound is not claimed to be tight.

Scenario	Geometric LB	Statistical LB	Combined LB	Empirical risk	Analytic risk	Risk / LB
d_star = 0.00, precise archive (s = 0.35)	0.000	0.012	0.012	0.099	0.099	8.524
d_star = 0.00, noisy archive (s = 1.20)	0.000	0.040	0.040	0.339	0.339	8.524
d_star = 0.25, precise archive (s = 0.35)	0.163	0.012	0.163	0.174	0.174	1.066
d_star = 0.25, noisy archive (s = 1.20)	0.163	0.040	0.163	0.365	0.363	2.236
d_star = 0.60, precise archive (s = 0.35)	0.391	0.012	0.391	0.392	0.392	1.001
d_star = 0.60, noisy archive (s = 1.20)	0.391	0.040	0.391	0.477	0.473	1.218
d_star = 1.00, precise archive (s = 0.35)	0.652	0.012	0.652	0.653	0.653	1.000
d_star = 1.00, noisy archive (s = 1.20)	0.652	0.040	0.652	0.674	0.675	1.033

When the hull distance is zero, increasing the common archive standard error from 0.35 to 1.20 increases the representative estimator's empirical worst-case MAE from 0.099 to 0.339, even though the geometric term vanishes. With precise archives and hull distances 0.60 and 1.00, the constructed lower bounds are 0.392 and 0.653, and the corresponding empirical worst-case MAEs are 0.392 and 0.653. This agreement illustrates the two restricted proof submodels. It neither numerically proves Theorem 5.5 nor establishes global minimax optimality of the representative estimator.

B.7 Bridge Design across Support Scenarios

The formal bridge experiment uses seeds 20261111–20261113, a 12-candidate library, budget four, bridge standard error 0.10, selection-error bound 0.01, three-point Gauss–Hermite quadrature, and 80 weight-optimization iterations per planning evaluation. At step b, policy π estimates

$$\widehat{\Delta}(u \mid S_{b-1}) = \widehat{\operatorname{diam}}\{\Theta_{\mathcal{A}(S_{b-1})}\} - \widehat{\mathbb{E}}\big[\operatorname{diam}\{\Theta_{\mathcal{A}(S_{b-1}\cup\{u\})}\} \mid \mathcal{A}(S_{b-1})\big]\,. \tag{B.7}$$

Causal greedy uses all four public representation coordinates, semantic greedy uses (s_1, s_2) only, and random selection is uniform. True mechanisms, target effects, and future bridge outcomes never enter planning. The experiment records planning and evaluation inconsistencies separately and never treats an empty intersection as a successful zero-diameter update. Table 16 is the formal 300-repetition bridge-policy comparison across support conditions.

Table 16: Formal bridge-design results across support conditions. Inconsistencies are recorded and never counted as zero diameter.

Scenario	Policy	Completion	Planning incons.	Initial diam.	Final diam.	Shrinkage	Selected bridges
Supported	Causal greedy	1.000	0.000	3.350	2.309	0.311	4.00
Supported	Semantic greedy	0.997	0.007	3.350	2.779	0.170	3.99
Supported	Random	1.000	0.000	3.350	2.637	0.213	4.00
Moderate	Causal greedy	1.000	0.000	3.787	2.067	0.454	4.00
Moderate	Semantic greedy	0.983	0.020	3.787	2.658	0.298	3.97
Moderate	Random	1.000	0.000	3.787	2.528	0.332	4.00
Strong	Causal greedy	1.000	0.000	5.504	1.862	0.662	4.00
Strong	Semantic greedy	0.943	0.060	5.504	2.483	0.549	3.89
Strong	Random	1.000	0.000	5.504	2.385	0.567	4.00
Severe	Causal greedy	1.000	0.000	6.942	1.877	0.730	4.00
Severe	Semantic greedy	0.967	0.037	6.942	2.331	0.664	3.93
Severe	Random	1.000	0.000	6.942	2.317	0.666	4.00

Causal-support greedy reduces mean diameter by 31.1%, 45.4%, 66.2%, and 73.0% in the supported, moderate, strong, and severe scenarios, respectively, and attains the smallest mean final diameter among the three policies in every scenario. Semantic-only planning produces an empty planning intersection on 0.7%–6.0% of paths, whereas causal-support greedy completes all four selections on every path. All policies retain nonempty intersections under the common full-representation evaluation, so these empty sets diagnose planning-representation inconsistency rather than final evaluation failure. As an evaluation-only mechanism check, the severe-scenario distance from the target to the expanded true-mechanism hull falls from 1.001 to 0.032 after causal-support selection; those latent coordinates never enter planning.

The separate exhaustive benchmark uses 30 repetitions and enumerates all $\binom{12}{b}$ subsets after outcomes are observed: 12, 66, and 220 subsets for budgets 1, 2, and 3. It is an ex-post comparator only. The separate evaluation-only exhaustive benchmark is reported in Table 17.

Table 17: Small-library exhaustive bridge benchmark. Exhaustive selection uses realized bridge outcomes and is evaluation-only.

Budget	Repeats	Subsets	Greedy diam.	Optimal diam.	Greedy / optimal value	Same set
1	30	12	2.693	2.676	0.996	0.667
2	30	66	2.131	2.028	0.978	0.333
3	30	220	1.932	1.865	0.986	0.433

The exhaustive table is an empirical efficiency check, not an estimate of the weak-submodularity constant and not a numerical proof of Theorem 5.6. The formal policy table remains the authoritative 300-repetition result; the focused budget-path data used in Figure 4b contains 90 paths per policy, or 270 policy paths in total.

B.8 NSW Construction, Stability, and Error Distributions

Source and preprocessing. The source is the 445-row Dehejia–Wahba NSW randomized job-training sample, with 185 treated and 260 control units. The committed file is `data/nsw_dw.dta`; its NBER source location and SHA-256 are recorded in the repository metadata. The outcome is 1978 earnings divided by 1000. We standardize age, education, Black and Hispanic indicators, marital status, no-degree status, and 1974 and 1975 earnings. The restricted representation contains age, education, race indicators, marital status, and no-degree status. The design-enriched representation adds 1974 and 1975 earnings, local overlap, and neighborhood radius. Causal ATLAS retrieves up to 24 candidates in the restricted coordinates and learns its simplex composition in the enriched coordinates, while the semantic baselines use only the restricted representation. This fixed split is an auditable stress-test design, not a claim that the enriched coordinates are known causal variables.

Local objects and holdout. For every anchor, the local object is formed from its 50 nearest neighbors subject to minimum treated and control support. Each object stores the treated-minus-control contrast, a Welch standard error, overlap score $4\widehat{p}(1-\widehat{p})$, and the root-mean-square standardized neighborhood radius. Deterministic pruning at the 95th percentile of center norm or neighborhood radius followed by farthest-point coverage leaves 112 local objects. Each of three seeds generates 20 splits with 28 held-out objects, giving 1,680 target evaluations. The method never reads the held-out contrast or standard error before prediction. Table 18 summarizes this fixed NSW construction and evaluation protocol, while Table 19 gives the separate held-out-split stability results.

Table 18: Fixed NSW local-object construction and evaluation protocol.

Quantity	Value	Quantity	Value
Individuals	445	Treated / control	185 / 260
Standardized covariates	8	Neighborhood size	50
Minimum treated / control	8 / 8	Local objects	112
Holdout objects per split	28	Splits per seed	20
Evaluation seeds	3	Total held-out targets	1,680
Maximum archive candidates	24	Release tolerance	3.3

Appendix B.8. NSW target-level certificate diagnostic

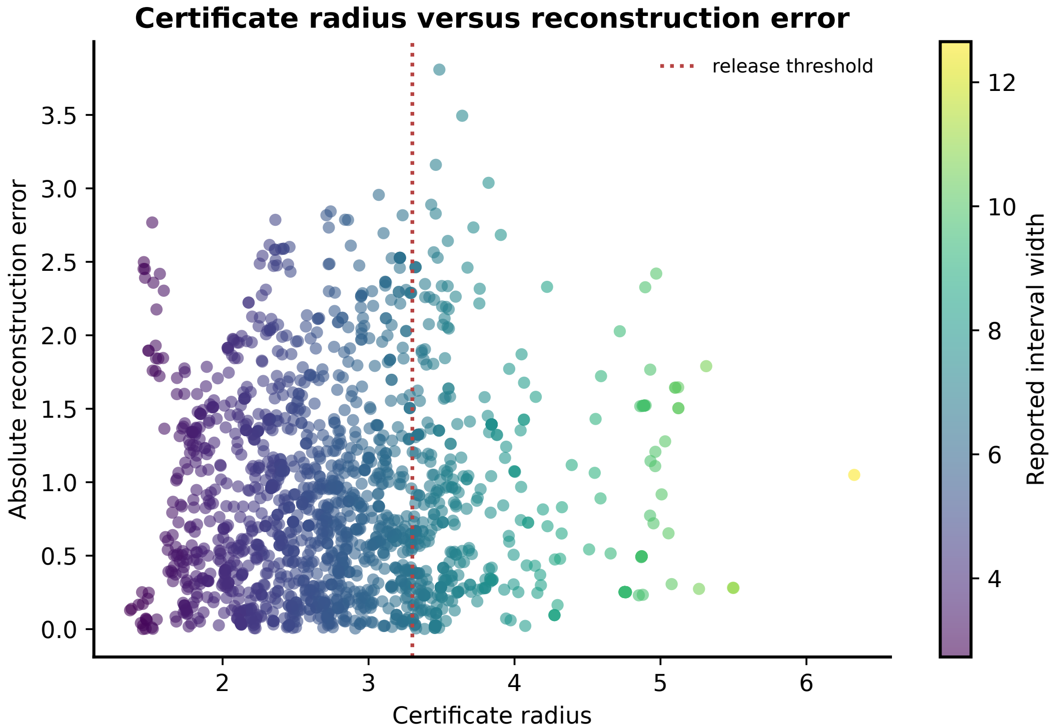

Figure 7: NSW certificate radius versus absolute reconstruction error across held-out local objects. Point colors encode reported interval width; the diagnostic is evaluation-only and uses noisy held-out contrasts.

Table 19: NSW reconstruction by held-out-split seed. Every row contains 560 target evaluations; monetary quantities are thousands of U.S. dollars.

Method	Seed	MAE	Median AE	Sign	Reference inclusion	Width	Rejection
Causal ATLAS	20261201	0.833	0.680	0.859	0.973	5.656	0.234
Causal ATLAS	20261202	0.854	0.702	0.839	0.970	5.587	0.232
Causal ATLAS	20261203	0.898	0.703	0.864	0.980	5.648	0.230
ATLAS, no rejection	20261201	0.833	0.680	0.859	0.970	5.299	0.000
ATLAS, no rejection	20261202	0.854	0.702	0.839	0.968	5.230	0.000
ATLAS, no rejection	20261203	0.898	0.703	0.864	0.971	5.295	0.000
Semantic forced	20261201	1.146	0.899	0.754	0.611	2.855	0.000
Semantic forced	20261202	1.162	0.926	0.745	0.620	2.855	0.000
Semantic forced	20261203	1.198	1.068	0.762	0.584	2.853	0.000
Nearest semantic	20261201	1.255	1.051	0.791	0.986	8.192	0.000
Nearest semantic	20261202	1.248	1.065	0.755	0.995	8.418	0.000
Nearest semantic	20261203	1.224	0.935	0.795	0.996	8.323	0.000
Global mean	20261201	1.640	1.387	0.795	0.436	2.473	0.000
Global mean	20261202	1.674	1.303	0.770	0.455	2.469	0.000
Global mean	20261203	1.723	1.337	0.787	0.446	2.481	0.000

The NSW certificate radius–reconstruction-error diagnostic is shown in Figure 7; it complements the seedwise summaries without changing their evaluation-only interpretation.

Point metrics and complete error distributions. ATLAS and its no-rejection counterpart have identical all-target MAE, median absolute error, and sign accuracy by construction: they use the same weights and raw predictions, and all 1,680 held-out objects are retained for point-error evaluation. Their difference is the release decision and the reported certificate interval, not the point estimator. Across the common holdouts, the absolute-error interquartile ranges are 0.297–1.291 for both ATLAS variants, 0.409–1.827 for semantic forced, 0.533–1.992 for nearest semantic, and 0.612–2.559 for the global mean. The corresponding 90th percentiles are 1.886, 2.464, 2.427, and 3.480. These distributional summaries show that the aggregate MAE ordering is not generated by a small number of extreme errors. The three seed batches have between-seed MAE standard deviations between 0.016 and 0.042 across methods, while the ATLAS rejection-rate standard deviation is 0.0018.

Held-out objects overlap in their underlying units, so their estimated contrasts are statistically dependent. The current protocol is intentionally descriptive and does not implement a full shared-unit covariance correction. In particular, reference inclusion is not called causal coverage, and the experiment does not support external-validity claims.

Reproducibility and Reporting Boundaries

All analyses are run from fixed scripts in the repository, and the committed result manifest records hashes for the source data, result tables, and paper assets. Seed-wise tables and target-level records remain available for audit but are not reproduced in full here. These experiments establish implementation behavior under the declared protocols, not universal performance. The synthetic archive has eight experiments, one smooth effect surface, known randomization, and oracle nuisance regressions; the negative controls intentionally violate certificate bounds. The NSW analysis uses noisy held-out contrasts and one fixed local-object construction. The numerical results should therefore be read as evidence for the framework's decision behavior and stated boundaries.

References

Bareinboim, E. and Pearl, J. (2016). Causal inference and the data-fusion problem. *Proceedings of the National Academy of Sciences.*

Chernozhukov, V., Chetverikov, D., Demirer, M., Duflo, E., Hansen, C., Newey, W., and Robins, J. (2018). Double/debiased machine learning for treatment and structural parameters. *The Econometrics Journal.*

Dehejia, R. H. and Wahba, S. (1999). Causal effects in nonexperimental studies: reevaluating the evaluation of training programs. *Journal of the American Statistical Association.*

Imbens, G. W. and Rubin, D. B. (2015). *Causal Inference for Statistics, Social, and Biomedical Sciences.* Cambridge University Press.

LaLonde, R. J. (1986). Evaluating the econometric evaluations of training programs with experimental data. *American Economic Review.*

Manski, C. F. (2003). *Partial Identification of Probability Distributions.* Springer.

Nemhauser, G. L., Wolsey, L. A., and Fisher, M. L. (1978). An analysis of approximations for maximizing submodular set functions. *Mathematical Programming.*

Neyman, J. (1923). On the application of probability theory to agricultural experiments. Essay on principles. Section 9. English translation in *Statistical Science*, 1990.

Pearl, J. (2009). *Causality*. Cambridge University Press.

Robins, J. M., Rotnitzky, A., and Zhao, L. P. (1994). Estimation of regression coefficients when some regressors are not always observed. *Journal of the American Statistical Association*.

Rubin, D. B. (1974). Estimating causal effects of treatments in randomized and nonrandomized studies. *Journal of Educational Psychology*.

van der Vaart, A. W. (1998). *Asymptotic Statistics*. Cambridge University Press.

Zhang, J., Bao, H., Wang, P., Yan, A., Liu, X., and Evans, J. A. (2026). Building an atlas of social experiments to link studies, reconcile conflicts, and bridge gaps. *arXiv preprint*.